QuaRK: A Resource-Aware Quantum Reservoir Kernel for Temporal Learning

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Abstract

Temporal learning requires representations that retain useful history while keeping statistical dependence and implementation cost explicit. We introduce QuaRK, a quantum-reservoir-based framework in which a finite-set Johnson–Lindenstrauss projection maps inputs to a quantum register, a bank of reset-contractive reservoirs processes each temporal window, local-observable expectations form a classical feature vector, and a regularized Matérn kernel provides the readout. The analysis follows this pipeline. Exact trace-norm contraction yields a unique causal state and a geometrically decaying finite-window approximation; Matérn RKHS geometry transfers this error to predictions and squared loss. Conditional on distinct exact features, the readout admits a finite-norm interpolant. For a fixed exact featurizer and stationary beta-mixing input–output data, an independent-block argument gives a one-sided risk bound governed by an explicit effective block count and the finite-window error. A finite-family extension makes this guarantee simultaneous over a predeclared collection of featurizer–readout configurations, adding only a logarithmic model-selection term. A separate classical-shadow result quantifies the measurements needed for local Pauli features and propagates finite-shot error uniformly over a bounded RKHS readout class. Together, these results provide a resource-aware, assumption-explicit basis for evaluating quantum reservoir kernels without claiming quantum advantage, hardware-noise robustness, or superiority over classical temporal models.

1 Introduction

Temporal learning is difficult for two related reasons. First, a prediction at one time may depend on events far in the past, so the learner needs a state that summarizes history without growing with the sequence length. Second, training examples extracted from a single trajectory are generally dependent, so their statistical behavior cannot be read directly from an i.i.d. analysis. Reservoir computing addresses the first difficulty by driving a fixed dynamical system with the input sequence and training only a comparatively simple readout (Jaeger, 2001; Maass et al., 2002). When the state dynamics are contractive, the effect of an arbitrary initialization vanishes, yielding a well-defined causal representation of the input history.

Quantum reservoir computing replaces the classical recurrent state by a quantum system and constructs classical features from measurements of the evolved state (Fujii & Nakajima, 2017; Chen et al., 2020). This offers a structured way to generate nonlinear temporal features while keeping the trainable component classical. It also makes the resource picture more layered than in an abstract reservoir model. Input dimension determines how many coordinates must be encoded, reservoir width and circuit depth determine the cost of one state evolution, the observable family determines the feature dimension, and finite-shot estimation determines how often the same temporal window must be replayed. These quantities should be analyzed separately rather than summarized by a single claim of scalability.

A second source of ambiguity is that several distinct questions are often compressed into one performance claim. A contractive state update can guarantee loss of initialization dependence without guaranteeing that different input histories produce different features. A positive-definite kernel can interpolate distinct feature vectors without establishing that the featurizer is injective. Likewise, an efficient estimator for many local observables does not by itself provide a learning guarantee under temporal dependence or a model of device noise. A useful theory for quantum reservoir learning should therefore connect these layers while preserving their individual assumptions.

We develop this perspective through QuaRK, a quantum-reservoir kernel for temporal learning. Each input vector is first mapped to the width of an n-qubit register by a finite-set Johnson–Lindenstrauss (JL) projection and a bounded angle encoding. A collection of independently parameterized reset-contractive quantum channels then processes a temporal window. Local observable expectations from the resulting reservoir states are concatenated into a classical feature vector, on which a regularized Matérn kernel readout is trained. The components have complementary roles: the projection controls encoded width, contraction controls the influence of the remote past, reservoir multiplexing and observable choice control feature dimension, and RKHS regularization controls the readout class.

The analysis follows the same order as the learning pipeline. We first account for the finite-set projection dimension, then use exact trace-norm contraction to construct a unique causal state and quantify the error introduced by a finite window. The geometry of the Matérn RKHS transfers this state-space error to predictions and squared loss. We next characterize the finite-sample capacity of the readout conditional on distinct exact features. Finally, an independent-block argument gives a one-sided risk bound for stationary beta-mixing input–output data. We extend this result from one fixed design to a predeclared finite family, so validation-based selection among finitely many reservoir and readout configurations incurs an explicit logarithmic penalty. A separate classical-shadow result quantifies the measurements needed to estimate local Pauli features. Keeping model selection and measurement separate from the exact-feature risk bound makes the scope of each result explicit.

Throughout the paper, *resource-aware* means that logical width, temporal channel applications, state preparations, observable locality, classical projection work, feature dimension, and kernel-training costs are reported as separate resources. It is not used as a synonym for computational advantage or practical efficiency.

Contributions. The main contributions are:

1. We formulate an end-to-end temporal learner built from reset-contractive quantum reservoirs and a classical Matérn kernel readout. Spatial multiplexing is represented by a tuple of independently evolved reservoir states followed by concatenated observable features, which keeps peak width and repeated execution costs distinct.

2. We prove exact trace-norm contraction of the reset dynamics, derive a unique causal state and a geometrically decaying finite-window feature error, and propagate this approximation through the Matérn RKHS to prediction and squared loss.

3. We separate readout capacity from statistical generalization. Conditional on pairwise-distinct exact features, strict positive definiteness of the Matérn kernel yields a finite-norm interpolant. For a fixed exact featurizer and stationary beta-mixing data, a corrected independent-block analysis yields a one-sided risk bound with an explicit effective block count and finite-window approximation term. A finite-family corollary makes the bound simultaneous over a predeclared collection of configurations and quantifies the additional model-selection cost.

4. We state the local-Pauli classical-shadow requirement independently of the learning bound and derive how coordinate-wise measurement error affects evaluation of a fixed RKHS readout.

The resulting contribution is an assumption-explicit foundation for studying quantum-reservoir-based temporal features. The theory does not establish quantum advantage or superiority over classical reservoirs; those are comparative empirical questions that require matched protocols.

Organization. After positioning the work in Section 2, we formalize temporal windows, reservoir features, risks, and dependence in Section 3. Section 4 then builds the complete learner and accounts for its resources. The theoretical analysis is developed in Section 5, with proofs in Appendix B. The empirical section then specifies the controlled functional hierarchy, dependent-family robustness study, high-dimensional width-control experiment, matched controls, and measurement ablations. The exact synthetic generators and task maps are collected in Appendices D and E, followed by the discussion and conclusion.

2 Related Work

QuaRK sits at the intersection of reservoir computing, quantum feature generation, kernel learning, and statistical learning from dependent data. The method does not replace the central ideas from these areas; rather, it combines them in a temporal architecture whose dynamical, statistical, and measurement assumptions can be tracked jointly.

Reservoir computing, causal filters, and fading memory. Echo-state networks and liquid-state machines established the principle of driving a fixed recurrent system and training only its readout (Jaeger, 2001; Maass et al., 2002). Fading memory formalizes continuity of a causal filter with respect to increasingly remote inputs (Boyd & Chua, 1985), while universality and risk analyses study, respectively, the functions representable by reservoir families and the statistical behavior of learned readouts (Grigoryeva & Ortega, 2018; Gonon et al., 2020). In the present work, uniform contraction is used for the narrower purpose of proving loss of initialization dependence and uniqueness of the causal state for one reset-based architecture. We do not rely on a separate universality or formal fading-memory theorem.

Quantum reservoir computing. Quantum reservoir computing realizes the driven state system with quantum dynamics (Fujii & Nakajima, 2017). Existing proposals have considered dissipative evolution, reset mechanisms, noisy processors, weak or repeated measurements, and spatial multiplexing (Chen & Nurdin, 2019; Chen et al., 2020; Nakajima et al., 2019; Molteni et al., 2023; Mujal et al., 2023; Hu et al., 2024). Finite-dimensional quantum reservoir filters and the role of input dependence have also received formal treatment (Martínez-Peña & Ortega, 2023). QuaRK uses a replacement channel whose contraction coefficient is exact and tunable. Its multiple reservoir instances are modeled as separate state evolutions, so they may be executed in parallel or sequentially on reused hardware; this distinction is important for separating peak width from runtime and state-preparation cost.

Quantum features and kernel readouts. Quantum feature methods encode classical data into quantum states or measurement statistics before applying a classical learner (Schuld & Killoran, 2019; Huang et al., 2021). Here the features are history-dependent: they are generated after a quantum reservoir has processed a temporal window. The downstream Matérn kernel is an ordinary classical positive-definite kernel on these measured features. Kernel ridge regression provides a regularized nonlinear readout, while the normalized RKHS structure bounds predictions and allows feature perturbations to be propagated to the readout (Genton, 2001; Porcu et al., 2024; Hofmann et al., 2008). Thus the quantum component lies in the temporal featurizer, not in a claim that the Matérn kernel lacks a classical analogue.

Projection and measurement. The JL lemma preserves pairwise geometry on a finite set under a random linear map whose target dimension grows logarithmically with the set size (Dasgupta & Gupta, 2003; Woodruff, 2014). We use it to relate the number of encoded coordinates to the finite collection of points being processed; the classical projection cost and the subsequent nonlinear dynamics remain part of the method. On the measurement side, classical shadows estimate many observables from shared randomized measurements (Huang et al., 2020). For local random Pauli measurements, a Pauli string of weight at most k has squared shadow norm scaling as 3^k, which determines the locality dependence in our shot bound.

Learning with dependent observations. Independent-block methods compare a beta-mixing sequence with a collection of independent blocks and yield uniform learning bounds in terms of an effective block count (Yu, 1994; Mohri & Rostamizadeh, 2008; Dedecker et al., 2007). We apply this machinery to strided temporal windows. The resulting confidence condition depends on the raw-time separation between blocks, and the stochastic terms scale with the number of approximately independent blocks rather than the raw number of windows.

The standard ingredients above become useful for QuaRK only after they are connected to the same temporal feature map. The technical contribution is therefore their architecture-specific integration: tuple-based multiplexing, finite-window error propagation through the Matérn readout, a window-process mixing calculation that retains the effective block count, and separate accounting of exact-feature learning and finite-shot measurement. We now introduce the stochastic objects needed to state this connection precisely.

3 Problem Setting

The analysis distinguishes three objects that coincide only in idealized settings: the causal feature of an infinite input history, the implementable feature obtained from a finite window, and the empirical sample of dependent windows used to train the readout. This section fixes these objects before the quantum architecture is specified.

3.1 Temporal data and windowed observations

Let $Z = (Z_t)_{t\in\mathbb{Z}}$, with $Z_t = (X_t, Y_t)$, be a stationary stochastic process on $\mathbb{R}^d \times \mathbb{R}$. We write $x_{-\infty:0} = (\ldots, x_{-1}, x_0)$ for a semi-infinite input history and

$$x_{-w+1:0} = (x_{-w+1}, \ldots, x_0) \in (\mathbb{R}^d)^w$$

for its most recent w inputs. A temporal predictor maps a history to a scalar. The target need not have finite memory; a finite window is instead an implementable approximation whose error will later be controlled by reservoir contraction.

For a window length $w \geq 1$ and stride $s = w + g$, with $g \geq 0$, define

$$W_j := \big(X_{js-w+1:js}, Y_{js}\big), \qquad j \in \mathbb{Z}. \tag{1}$$

The observed sample is $\mathcal{D}_N = (W_1, \ldots, W_N)$. The gap g controls the raw-time separation between consecutive windows, but nonoverlap alone does not make them independent.

3.2 Causal and finite-window reservoir features

There are R independently parameterized sub-reservoirs. Sub-reservoir r has an n-qubit state $\rho_t^{(r)}$ and an input-dependent channel $T_{r,x}$. For a semi-infinite history, its causal state at time zero is written $\rho_{\infty,0}^{(r)}(x_{-\infty:0})$. For a finite window, the recursion starts at time $-w$ from a fixed state $\rho_{\text{init}}^{(r)}$, producing $\rho_{w,0}^{(r)}(x_{-w+1:0})$ after the last input is consumed.

Let $\mathcal{O}$ be a finite family of Hermitian observables with $\|O\|_\infty \leq 1$. The exact infinite-history and finite-window feature maps are

$$\Phi_\infty(x_{-\infty:0}) := \big(\mathrm{Tr}[O\rho_{\infty,0}^{(r)}]\big)_{r\in[R],\, O\in\mathcal{O}}, \tag{2}$$

$$\Phi_w(x_{-w+1:0}) := \big(\mathrm{Tr}[O\rho_{w,0}^{(r)}]\big)_{r\in[R],\, O\in\mathcal{O}} \in \mathbb{R}^p, \qquad p = R|\mathcal{O}|. \tag{3}$$

These vectors concatenate expectations from a tuple of independently evolved states; no tensor-product state is introduced for multiplexing. A finite-shot estimate $\widehat{\Phi}_w$ is a distinct random object and will be treated separately from the exact feature map.

3.3 Kernel readout and the three risks

Let $\kappa : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ be a normalized Matérn kernel and $\mathcal{H}_\kappa$ its RKHS. For a radius $\Lambda > 0$, define

$$\mathcal{H}_{\kappa,\Lambda} := \left\{ h \in \mathcal{H}_\kappa : \|h\|_{\mathcal{H}_\kappa} \leq \Lambda \right\}.$$

Normalization gives $\kappa(z, z) = 1$, and hence

$$|h(z)| \leq \|h\|_{\mathcal{H}_\kappa} \sqrt{\kappa(z, z)} \leq \Lambda, \qquad h \in \mathcal{H}_{\kappa,\Lambda}. \tag{4}$$

For squared loss $\ell(\widehat{y}, y) = (\widehat{y} - y)^2$, the causal target is the infinite-history risk

$$\mathcal{R}_\infty(h) := \mathbb{E}\big[\ell\big(h(\Phi_\infty(X_{-\infty:0})), Y_0\big)\big] . \tag{5}$$

The finite-window approximation is

$$\mathcal{R}_w(h) := \mathbb{E}\big[\ell\big(h(\Phi_w(X_{-w+1:0})), Y_0\big)\big] , \tag{6}$$

and the quantity optimized from the observed windows is

$$\widehat{\mathcal{R}}_{N,w}(h) := \frac{1}{N}\sum_{j=1}^{N} \ell\big(h(\Phi_w(X_{js-w+1:js})), Y_{js}\big). \tag{7}$$

The learning analysis will therefore have two parts: a statistical comparison between $\mathcal{R}_w$ and $\widehat{\mathcal{R}}_{N,w}$, and a deterministic comparison between $\mathcal{R}_\infty$ and $\mathcal{R}_w$.

3.4 Dependence and analytical scope

For sub-sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, let $\beta(\mathcal{A}, \mathcal{B})$ denote their absolute-regularity coefficient. We use

$$\beta_Z(q) := \sup_{t\in\mathbb{Z}} \beta(\sigma(Z_u : u \le t), \sigma(Z_u : u \ge t+q)) , \qquad q \ge 0, \tag{8}$$

and call the process beta mixing when $\beta_Z(q) \to 0$.

The primary learning result is first stated for a fixed design. The projection, reservoir parameters, initialization states, observable family, window scheme, and Matérn parameters are chosen independently of the sample on which that result is evaluated; when specialized to kernel ridge regression, its regularization parameter is fixed on the same basis. Random design elements, such as the JL matrix or reservoir seed, may be drawn independently and then conditioned upon. We assume that $x \mapsto V_r(x)$, equivalently $x \mapsto T_{r,x}$, is measurable, that the resulting loss classes are measurable (or admit the usual separable versions), that $|Y_t| \le \Upsilon$ almost surely for some $\Upsilon > 0$, and that the Matérn kernel is normalized. Results using its spectral second moment require $\nu > 1$. The main risk bound concerns exact observable expectations and is uniform over $\mathcal{H}_{\kappa,\Lambda}$. Corollary 10 later extends this uniformity to sample-dependent selection from a finite family of configurations that is fully declared before the sample is examined, including all random seeds and preprocessing maps. Continuous or adaptively generated hyperparameter families remain outside that extension.

With the data, feature maps, risks, and dependence structure in place, we next specify the concrete QuaRK pipeline that produces Φ_w.

4 Method

QuaRK turns a temporal window into a kernel input through four stages. The raw observations are first projected and encoded as rotations. Each projected sequence then drives several reset-contractive quantum reservoirs. Local observable expectations from their final states are concatenated, and a Matérn kernel ridge-regression readout is trained on the resulting vectors. We describe the stages in this order and conclude with an end-to-end resource ledger.

4.1 Finite-set projection and angle encoding

For resource accounting, let $\mathcal{P} \subset \mathbb{R}^d$ be the set of all individual time-point vectors appearing in the finite collection of windows to be processed, so $|\mathcal{P}| \le wN$. Independently of these points, and for a chosen register width n, draw

$$\Pi \in \mathbb{R}^{n\times d}, \qquad \Pi_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n),$$

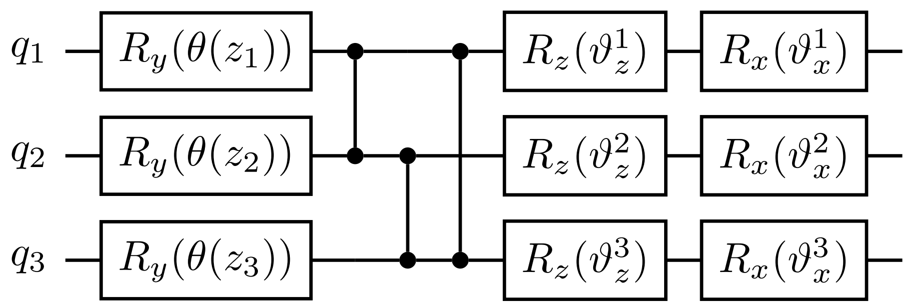

Figure 1: Example input-dependent unitary for a three-qubit sub-reservoir. The controlled links indicate a generic topology-constrained entangling pattern.

and set $z = \Pi x$. Each coordinate is converted to a bounded angle, for example $\theta(z_j) = \pi \tanh(z_j)$, and encoded by a one-qubit $R_y(\theta(z_j))$ rotation.

The finite-set JL statement in proposition 1 allows the encoded dimension to be chosen from a distortion and failure-probability target rather than by setting $n = d$. This width interpretation applies only to the projected finite set: the dense multiplication still costs $O(nd)$ operations per time point, and the subsequent angle encoding and reservoir evolution are nonlinear. The encoded rotations now provide the input layer of each reservoir update.

4.2 Reset-contractive quantum dynamics

Sub-reservoir r applies the input-dependent unitary

$$V_r(x) = W_r \bigotimes_{j=1}^{n} R_y(\theta((\Pi x)_j))\,, \tag{9}$$

where W_r is a fixed topology-constrained unitary. Its parameters are drawn once and then held fixed. A generic three-qubit instance is shown in Figure 1; the controlled links represent a chosen entangling pattern rather than a particular device-native gate set.

After the unitary, the state is partially replaced by a fixed reset state. On an arbitrary operator A, define

$$T_{r,x}(A) := \lambda_r V_r(x) A V_r(x)^\dagger + (1-\lambda_r)\,\mathrm{Tr}[A]\,\rho_{\mathrm{reset}}, \qquad 0 \le \lambda_r < 1. \tag{10}$$

The map $A \mapsto \mathrm{Tr}[A]\rho_{\mathrm{reset}}$ is a replacement channel. A Kraus representation is given in Appendix A, so it is completely positive and trace preserving; consequently, so is $T_{r,x}$. For a density operator ρ, this reduces to

$$T_{r,x}(\rho) = \lambda_r V_r(x) \rho V_r(x)^\dagger + (1-\lambda_r)\rho_{\mathrm{reset}}.$$

The parameter λ_r sets the worst-case memory scale: values near one retain more of the previous state, whereas smaller values place more weight on the reset state.

The replacement channel admits the ideal dilation in Figure 2. For $\rho_{\mathrm{reset}} = |+\rangle\langle+|^{\otimes n}$, an n-qubit reset register is prepared in $|+\rangle^{\otimes n}$, and a coin qubit controls swaps between the reservoir and reset registers. Discarding the coin and reset registers produces the mixture in Equation (10).

For a length-w window, the recursion starts from $\rho_{\mathrm{init}}^{(r)}$ and applies Equation (10) once per time point. Moving the initialization into the remote past defines the causal state; lemma 2 will show that this limit exists and is unique.

4.3 Reservoir multiplexing and observable features

The R sub-reservoirs produce a tuple of final states,

$$\big(\rho_{w,0}^{(1)}, \ldots, \rho_{w,0}^{(R)}\big),$$

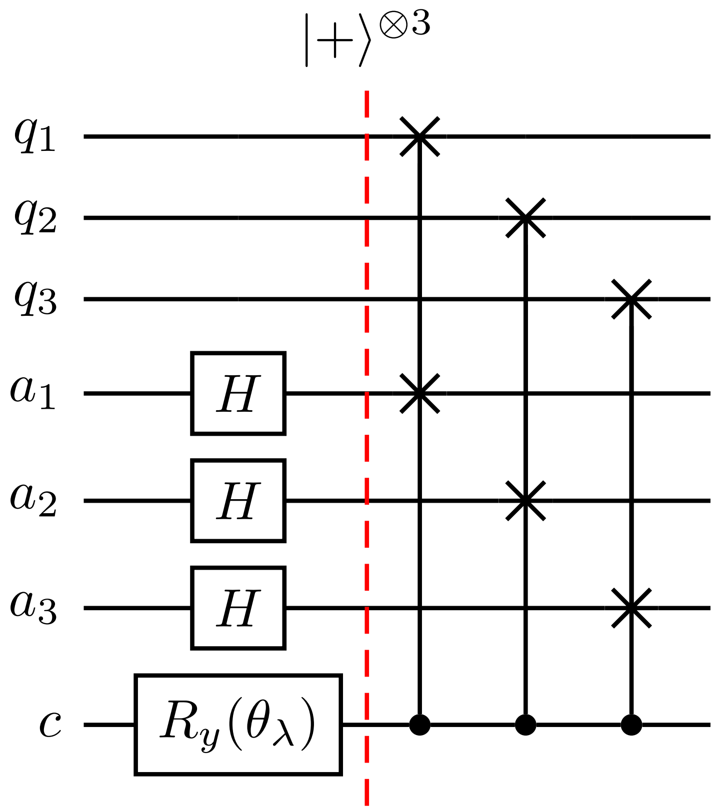

Figure 2: Ideal dilation of the reset channel for three reservoir qubits, with $\theta_\lambda = 2\arcsin\sqrt{1-\lambda}$. The reduced channel is obtained by discarding the coin and reset registers.

rather than a tensor-product state. They may be run in parallel on separate registers or sequentially on reused hardware. Parallel execution multiplies peak quantum width, whereas sequential execution keeps the reservoir width at n qubits but multiplies runtime and state preparations.

For each reservoir, the observables in $\mathcal{O}$ are evaluated and concatenated as in Equation (3), producing a vector of dimension $p = R|\mathcal{O}|$. The exact expectations define the analytic feature map used in the kernel and in the main learning theorem. Their finite-shot estimates are introduced later in proposition 12 and corollary 13. Once the quantum states have been reduced to this classical vector, the remaining learning stage is an ordinary kernel method.

4.4 Matérn kernel readout

For distance $r \geq 0$, smoothness $\nu > 1$, and length scale $\xi > 0$, let

$$\varphi_{\nu,\xi}(r) := \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\, r}{\xi} \right)^{\nu} K_\nu \left(\frac{\sqrt{2\nu}\, r}{\xi} \right), \tag{11}$$

where K_ν is the modified Bessel function of the second kind. With $\varphi_{\nu,\xi}(0) = 1$, define

$$\kappa(z, z') := \varphi_{\nu,\xi}(\|z - z'\|_2) \,. \tag{12}$$

This is a classical kernel evaluated after feature extraction. We use its strict positive definiteness to study interpolation and its spectral second moment to control sensitivity to feature perturbations.

Table 1: Principal resource parameters of QuaRK.

Quantity	Accounting
Encoded reservoir width	n qubits per sub-reservoir
Ideal reset dilation	Up to $2n+1$ qubits when reset ancillas and coin are simultaneous
Reservoir realizations	R; sequential reuse or R-fold parallel width
Exact feature dimension	$p = R\|\mathcal{O}\|$
Projection work	$O(nd)$ per time point for a dense Π; values may be cached across snapshots
Per-state temporal evolution	w channel applications, each containing input encoding, W_r, and one reset-channel realization
Finite-shot state preparations	NRS for N windows, R reservoirs, and S snapshots per state
Finite-shot channel updates	$NRSw$; gate count and depth depend on W_r and the reset implementation
Kernel storage/training	$O(N^2)$ memory and $O(N^3)$ naive time

Given exact training features $z_i = \Phi_w(x_i)$, kernel ridge regression solves

$$\min_{h\in\mathcal{H}_\kappa} \frac{1}{N}\sum_{i=1}^{N}(h(z_i)-y_i)^2 + \eta\,\|h\|_{\mathcal{H}_\kappa}^2 . \tag{13}$$

By the representer theorem (Kimeldorf & Wahba, 1971; Schölkopf & Smola, 2002), $\widehat{h}_\eta(\cdot) = \sum_i \alpha_i \kappa(z_i,\cdot)$, where

$$\boldsymbol{\alpha} = (K + N\eta I)^{-1}\boldsymbol{y}.$$

When $|y_i| \le \Upsilon$, comparing the objective at $\widehat{h}_\eta$ with the zero function gives

$$\eta\left\|\widehat{h}_\eta\right\|_{\mathcal{H}_\kappa}^2 \le \frac{1}{N}\sum_{i=1}^{N} y_i^2 \le \Upsilon^2, \qquad \left\|\widehat{h}_\eta\right\|_{\mathcal{H}_\kappa} \le \frac{\Upsilon}{\sqrt{\eta}}. \tag{14}$$

Thus fixed-η KRR belongs to the RKHS ball used in Section 5 with $\Lambda = \Upsilon/\sqrt{\eta}$, provided the featurizer, kernel parameters, and η are fixed independently of the analyzed sample. The finite-family result in corollary 10 also allows these quantities to be selected from a predeclared finite grid. Dense KRR requires $O(N^2)$ kernel storage and $O(N^3)$ naive linear-algebra time, which remains part of the end-to-end cost.

4.5 Resource summary

The stages above expose different bottlenecks. Table 1 separates peak quantum width from repeated temporal evolution and from classical post-processing. In particular, one snapshot of a final length-w reservoir state requires replaying all w channel updates. Projected inputs may be cached, but the quantum evolution must be repeated for each state preparation.

This decomposition will reappear in the theory: the projection controls encoded width, contraction controls the finite-history approximation, RKHS geometry controls the readout, dependence determines the effective sample size, and observable locality controls the measurement budget.

5 Theory

The method naturally leads to a sequence of analytical questions. How wide must the encoded input be on a finite dataset? Does the reservoir state depend on an arbitrary initialization? How accurately does a finite window represent the causal infinite-history feature? How does this approximation interact with the Matérn readout, how does temporal dependence change the learning rate, and what price is paid when validation selects among finitely many candidate configurations? We answer these questions in the same order as the pipeline and then return to the finite-shot cost of estimating the observable features. Unless stated otherwise, the results use the analytical scope specified in Section 3.4. Complete proofs are collected in Appendix B.

5.1 Finite-set projection

Proposition 1 (Finite-set JL projection; adapted from Dasgupta & Gupta, 2003; Woodruff, 2014)**.** *Let $\mathcal{P} \subset \mathbb{R}^d$ contain $M \geq 2$ points. For $\varepsilon, \delta \in (0,1)$, there is a universal constant $C > 0$ such that, if*

$$n \geq C\varepsilon^{-2} \log\left(\frac{M}{\delta}\right)$$

and $\Pi_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n)$, then with probability at least $1-\delta$,

$$(1-\varepsilon)\left\|u-v\right\|_2^2 \leq \left\|\Pi u - \Pi v\right\|_2^2 \leq (1+\varepsilon)\left\|u-v\right\|_2^2$$

simultaneously for all $u, v \in \mathcal{P}$. For the points occurring in N windows of length w, one may take $M \leq wN$.

For a prescribed finite set and distortion level, the proposition makes the encoded dimension logarithmic in the number of points rather than tied directly to d. If the point set is the realized sample, the statement is read conditionally on that set, with Π drawn independently. This JL event is separate from the learning-theorem event unless their failure probabilities are combined explicitly. The statement concerns the projection before angle encoding: it neither guarantees that the final reservoir features are distinct nor removes the d-dependent cost of applying Π. Having accounted for encoded width, we next study the temporal state generated from these inputs.

5.2 Contraction, causal states, and finite windows

Lemma 2 (Exact trace-norm contraction)**.** *For every sub-reservoir r, input x, and pair of density operators ρ, ρ',*

$$\left\|T_{r,x}(\rho) - T_{r,x}(\rho')\right\|_1 = \lambda_r \left\|\rho - \rho'\right\|_1. \tag{15}$$

Consequently, for each input history there is a unique causal state obtained as the initialization time tends to $-\infty$. Under the measurability condition in Section 3.4, the causal-state map is measurable. Two trajectories driven by the same last w inputs satisfy

$$\left\|\rho_0^{(r)} - \rho_0'^{(r)}\right\|_1 \leq 2\lambda_r^w. \tag{16}$$

The state-level estimate immediately controls the observable representation used by the learner.

Proposition 3 (Finite-window feature error)**.** *For every input history,*

$$\begin{aligned}\left\|\Phi_\infty(x_{-\infty:0}) - \Phi_w(x_{-w+1:0})\right\|_2 &\leq 2\sqrt{|\mathcal{O}| \sum_{r=1}^{R} \lambda_r^{2w}} && (17)\\ &\leq 2\sqrt{R|\mathcal{O}|}\,\lambda_{\max}^w, \qquad \lambda_{\max} := \max_r \lambda_r. && (18)\end{aligned}$$

Together, lemma 2 and proposition 3 show that the reservoir has a unique input-determined causal state and that a finite-window feature vector approaches its infinite-history counterpart geometrically. The parameters λ_r therefore expose a worst-case memory scale, while the window length w controls the corresponding approximation error. The next step is to determine how this feature error affects the kernel readout.

5.3 Matérn stability and truncation of loss

Lemma 4 (Matérn RKHS section stability; derived from Genton, 2001; Porcu et al., 2024)**.** *Let κ be the normalized Matérn kernel in Equations (11) and (12) with $\nu > 1$. Then*

$$\|\kappa(z, \cdot) - \kappa(z', \cdot)\|_{\mathcal{H}_\kappa} \leq \frac{1}{\xi}\sqrt{\frac{\nu}{\nu - 1}}\, \|z - z'\|_2 \,. \tag{19}$$

Corollary 5 (Prediction and loss truncation)**.** *For every $h \in \mathcal{H}_{\kappa,\Lambda}$,*

$$|h(\Phi_\infty) - h(\Phi_w)| \leq \frac{2\Lambda}{\xi}\sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1}\sum_{r=1}^{R} \lambda_r^{2w}}, \tag{20}$$

$$|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \leq \tau_w, \tag{21}$$

where

$$\tau_w := \frac{4\Lambda(\Lambda + \Upsilon)}{\xi}\sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1}\sum_{r=1}^{R} \lambda_r^{2w}}. \tag{22}$$

The corollary converts dynamical forgetting into a uniform approximation of predictions and squared risk over the RKHS ball. This term will appear additively in the learning bound. In particular, increasing the sample size alone cannot remove τ_w when w and the reservoir parameters are held fixed.

5.4 Conditional interpolation capacity

The truncation result concerns stability of a bounded readout. A separate question is how much norm is needed to fit a finite set of exact features.

Theorem 6 (Conditional finite-sample interpolation; cf. Aronszajn, 1950; Kimeldorf & Wahba, 1971; Hofmann et al., 2008)**.** *Let $z_i = \Phi_w(x_i) \in \mathbb{R}^p$, $i \in [N]$, be pairwise distinct exact feature vectors, and let $K_{ij} = \kappa(z_i, z_j)$. The Matérn Gram matrix K is positive definite. If $\boldsymbol{y} \neq 0$, define*

$$\Lambda_\star := \sqrt{\boldsymbol{y}^\top K^{-1} \boldsymbol{y}}.$$

Then there is a unique minimum-norm interpolant $h_\star \in \mathcal{H}_\kappa$ with $\|h_\star\|_{\mathcal{H}_\kappa} = \Lambda_\star$ and $h_\star(z_i) = y_i$ for all i. Therefore:

1. *for $\Lambda \geq \Lambda_\star$,*
$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) = 0;$$

2. *for $0 \leq \Lambda \leq \Lambda_\star$,*
$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) \leq \left(1 - \frac{\Lambda}{\Lambda_\star}\right)^2 \frac{\|\boldsymbol{y}\|_2^2}{N}. \tag{23}$$

If $\boldsymbol{y} = 0$, the zero function interpolates with norm zero.

The theorem isolates the capacity of the Matérn readout: once the exact feature vectors are distinct, a finite RKHS norm suffices to interpolate them, and scaling the interpolant gives a constructive training-error bound below that norm. Pairwise distinctness is an assumption of the result; neither the JL projection nor reservoir contraction guarantees it. This distinction is important because interpolation capacity alone says nothing about test risk. We now turn to that statistical question.

5.5 Dependence of strided windows

The learning sample is the window process (W_j), not the raw process (Z_t). The following lemma translates their mixing coefficients by tracking the first raw-time index appearing in a future window.

Lemma 7 (Mixing coefficient of the window process)**.** *Let W be the process in Equation (1) with stride $s = w + g$. For every integer $a \geq 1$,*

$$\beta_W(a) \leq \beta_Z(as - w). \tag{24}$$

In particular, $\beta_W(1) \leq \beta_Z(g)$.

The first raw input in a window at lag a occurs at $as - w + 1$. The displayed bound is conservative by one time index under Equation (8), using monotonicity of beta-mixing coefficients. It supplies the dependence term needed by the independent-block construction.

5.6 Generalization with dependent windows

For a sample $\mathcal{S}_\mu = (\widetilde{W}_1, \ldots, \widetilde{W}_\mu)$, define the empirical Rademacher complexity using the convention

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) := \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{f \in \mathcal{F}} \left| \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\widetilde{W}_j) \right| \right], \quad \sigma_j \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}\{-1, +1\}. \tag{25}$$

This is the absolute-value convention used by Mohri & Rostamizadeh (2008); keeping it explicit avoids hidden factor or centering changes when their theorem is specialized. For a length $N = 2\mu a$ sequence of windows, $\mathcal{S}_\mu$ denotes one fixed position from each of the μ odd blocks of length a. Let

$$\mathcal{F}_\Lambda := \left\{ (x_{-w+1:0}, y) \mapsto (h(\Phi_w(x_{-w+1:0})) - y)^2 : h \in \mathcal{H}_{\kappa,\Lambda} \right\}.$$

Theorem 8 (One-sided risk bound for beta-mixing windows; adapted from Mohri & Rostamizadeh, 2008)**.** *Under the analytical scope of Section 3.4, suppose that Z is stationary and beta mixing. Let $N = 2\mu a$, with $\mu, a \in \mathbb{N}$, and set*

$$\delta' := \delta - 4(\mu - 1)\beta_Z(as - w). \tag{26}$$

For every $\delta \in (0, 1)$ satisfying $\delta' > 0$, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}_{\kappa,\Lambda}$,

$$\mathcal{R}_\infty(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} + \tau_w. \tag{27}$$

Define the source-convention squared-loss complexity constant

$$C_{\Lambda,\Upsilon} := 8\Lambda(\Lambda + \Upsilon) + 2\Upsilon^2. \tag{28}$$

Corollary 9 (Closed-form RKHS bound; using the contraction principle of Bartlett & Mendelson, 2002)**.** *Under the conditions of theorem 8, with the same probability, simultaneously for all $h \in \mathcal{H}_{\kappa,\Lambda}$,*

$$\begin{aligned} \mathcal{R}_\infty(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \frac{C_{\Lambda,\Upsilon}}{\sqrt{\mu}} + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} \\ + \frac{4\Lambda(\Lambda + \Upsilon)}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1} \sum_{r=1}^{R} \lambda_r^{2w}}. \end{aligned} \tag{29}$$

The bound decomposes causal true risk into finite-window training risk, two stochastic terms based on the effective block count $\mu = N/(2a)$, and the deterministic truncation term τ_w. Its uniformity over the fixed RKHS ball includes fixed-η KRR through Equation (14), with $\Lambda = \Upsilon/\sqrt{\eta}$. The statement is one-sided, and the condition $\delta' > 0$ makes the dependence–sample-size trade-off explicit: holding $a = 1$ while increasing N need not preserve the confidence condition.

5.7 Selection from a finite candidate family

The preceding bound fixes the featurizer and kernel before observing the sample. Empirical studies, however, commonly compare a finite grid of reservoir and readout configurations and select one using validation performance. The next result makes the bound simultaneous over such a predeclared family.

Corollary 10 (Finite-family model selection; finite-union argument based on McDiarmid, 1989; Bartlett & Mendelson, 2002)**.** *Let $\mathfrak{C} = \{1, \ldots, L\}$, with $L \geq 1$, be a finite, measurable family fixed before observing the sample, including every candidate's random seed and preprocessing map. All candidates share the same window length w, stride s, and block parameters $N = 2\mu a$. Candidate ℓ specifies an exact causal and finite-window featurizer $\Phi_\infty^{(\ell)}, \Phi_w^{(\ell)}$, a normalized Matérn kernel κ_ℓ with $\nu_\ell > 1$, and an RKHS ball*

$$\mathcal{H}_\ell := \left\{ h \in \mathcal{H}_{\kappa_\ell} : \|h\|_{\mathcal{H}_{\kappa_\ell}} \leq \Lambda_\ell \right\}.$$

Let $\tau_{w,\ell}$ be its truncation term from corollary 5, define

$$B_{\mathfrak{C}} := \max_{\ell \in [L]} (\Lambda_\ell + \Upsilon)^2, \qquad A_{\mathfrak{C}} := \max_{\ell \in [L]} \left[8\Lambda_\ell(\Lambda_\ell + \Upsilon) + 2\Upsilon^2 \right],$$

and retain $\delta' = \delta - 4(\mu - 1)\beta_Z(as - w) > 0$. Then, with probability at least $1 - \delta$, simultaneously for every $\ell \in [L]$ and $h \in \mathcal{H}_\ell$,

$$\begin{aligned} \mathcal{R}_{\infty,\ell}(h) \leq \widehat{\mathcal{R}}_{N,w,\ell}(h) + \frac{A_{\mathfrak{C}}}{\sqrt{\mu}} + 2B_{\mathfrak{C}} \sqrt{\frac{2 \log L}{\mu}} \\ + 3B_{\mathfrak{C}} \sqrt{\frac{\log(4/\delta')}{2\mu}} + \tau_{w,\ell}. \end{aligned} \tag{30}$$

Here $\mathcal{R}_{\infty,\ell}$ and $\widehat{\mathcal{R}}_{N,w,\ell}$ denote the risks induced by candidate ℓ. Consequently, the same event controls any measurable, sample-dependent selector $(\widehat{\ell}, \widehat{h})$ whose output belongs to $\bigcup_{\ell=1}^{L} \{\ell\} \times \mathcal{H}_\ell$.

The additional term scales as $\sqrt{\log L/\mu}$, so validation-based selection among a declared finite grid costs only logarithmically in the number of candidates. The result can cover finite choices of reservoir parameters, random seeds, observable families, and Matérn readout parameters, provided the candidate list and its preprocessing are fixed in advance and the candidates share the window scheme used by the blocking argument. It does not cover continuous optimization, adaptive expansion of the search space, or sample-estimated preprocessing that is not included in the finite family.

For geometric mixing, the block length can be increased slowly enough to control the coupling penalty without reducing the effective sample size too severely.

Corollary 11 (Geometrically mixing case)**.** *Suppose, in addition, that $\beta_Z(q) \leq \beta_0 e^{-cq}$ for constants $\beta_0, c > 0$, and keep w and s fixed. For a compatible sample size $N = 2\mu a$, choose an integer $a \leq N/2$ satisfying*

$$a \geq \left\lceil \max\left\{ 1, \, \frac{1}{cs} \log\left(\frac{4\beta_0 e^{cw} N}{\delta} \right) \right\} \right\rceil. \tag{31}$$

Then $\delta' \geq \delta/2$. For fixed $\Lambda, \Upsilon, \delta, \beta_0, c, w, s$, the two stochastic terms in Equation (29) are

$$O\left(\sqrt{\frac{\log(N/\delta)}{N}} \right).$$

If the available sample size is not divisible by $2a$, retain $N_a = 2a\lfloor N/(2a) \rfloor$ observations (provided $N_a \geq 2a$); at most $2a-1$ terminal observations are discarded. For fixed problem parameters, the displayed choice satisfies $a \leq N/2$ for all sufficiently large N.

The corollary recovers an independent-like rate up to a logarithmic factor, but it does not remove the finite-window term. A consistency result would require a joint schedule for w, s, a, and the readout class.

5.8 Finite-shot local-observable estimation

The preceding learning bound treats Φ_w as an exact expectation vector. On a measurement-based implementation, every coordinate must instead be estimated from repeated preparations of the final reservoir state.

Proposition 12 (Local-Pauli classical-shadow requirement; adapted from Huang et al., 2020)**.** *Suppose $\mathcal{O}$ contains $M = |\mathcal{O}|$ nonidentity Pauli strings of weight at most k. In each snapshot, measure every qubit in an independently and uniformly selected X, Y, or Z basis. There is a universal constant $C_{\mathrm{sh}} > 0$ such that*

$$S \geq C_{\mathrm{sh}}\, 3^k \varepsilon^{-2} \log\left(\frac{2M}{\delta_{\mathrm{sh}}}\right) \tag{32}$$

snapshots of one reservoir state suffice for simultaneous additive error at most ε for all observables, with probability at least $1 - \delta_{\mathrm{sh}}$.

For N windows and R separately prepared sub-reservoir states, the choice

$$S \geq C_{\mathrm{sh}}\, 3^k \varepsilon^{-2} \log\left(\frac{2MNR}{\delta_{\mathrm{tot}}}\right) \tag{33}$$

gives the same coordinate-wise error simultaneously over all NR states, by a union bound. This requires NRS state preparations and $NRSw$ temporal channel applications. Identity observables, if retained in a feature ledger, are known exactly.

Corollary 13 (Uniform stability of RKHS readouts to measurement error)**.** *Suppose every coordinate of $\widehat{\Phi}_w - \Phi_w$ has magnitude at most ε. On this deterministic event, the following inequalities hold simultaneously for every $h \in \mathcal{H}_{\kappa,\Lambda}$:*

$$\left| h(\widehat{\Phi}_w) - h(\Phi_w) \right| \leq \frac{\Lambda}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}}\, \varepsilon, \tag{34}$$

$$\left| \ell(h(\widehat{\Phi}_w), y) - \ell(h(\Phi_w), y) \right| \leq \frac{2\Lambda(\Lambda + \Upsilon)}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}}\, \varepsilon \tag{35}$$

whenever $|y| \leq \Upsilon$.

For the stated measurement ensemble, observable locality determines the exponential factor 3^k, while the number of observables enters only logarithmically. The perturbation corollary is uniform over the RKHS ball and therefore applies to any already-chosen readout in that ball. It compares the same function on exact and estimated features; it does not compare a readout retrained on noisy features with the exact-feature estimator, nor does it cover physical errors in state preparation and evolution. Those effects are separate from finite-shot sampling.

6 Empirical Evaluation

The empirical study is a predeclared behavioral characterization of QuaRK. It is not designed to find one favorable benchmark or to support a quantum-advantage claim. Instead, it asks when the method succeeds, when it fails, and how its behavior changes with the target functional, the dependence mechanism, the ambient dimension, the reservoir resources, and the measurement implementation. Experiments E1–E6 use controlled stochastic processes; E7 deliberately moves outside the assumptions of the learning theory; E8 is a limited external sanity check; and E9 is a compact IBM hardware-feasibility study. The exact run matrices, seed policy, readout grid, metrics, artifact identifiers, and failure rules are fixed in Appendix C. The data generators and targets are defined in Appendices D and E, with the out-of-scope and external protocols in Appendices F to H.

6.1 Empirical questions

Q1: Which temporal computations are recoverable from the final reservoir features? E1 moves from genuine one-step prediction to distributed linear memory, second-order memory, delayed recall, and a sparse cross-lag interaction. The regularization path and nested-sample curves diagnose interpolation, conditioning, and finite-sample behavior without fitting a prespecified asymptotic rate.

Q2: How do process persistence, task memory, and reservoir contraction interact? E2 varies the process persistence and the target delay separately from the reset rate. It tests whether a longer-memory operating point is useful when history is informative and whether excessive persistence can become harmful. It then checks whether the resulting strengths and failures transfer from VARMA to latent-regime and volatility-clustering processes.

Q3: What remains when the ambient dimension grows but encoded width is fixed? E3 distinguishes sparse signal plus nuisance coordinates from signal spread across all coordinates. It reports prediction, empirical JL distortion, projection cost, and width sensitivity. This can support a fixed-encoded-width claim for the evaluated construction, but not end-to-end cost independence from the original dimension.

Q4: How does QuaRK compare with classical representations under transparent matching rules? E4 consolidates, rather than reruns, selected scenarios from E1–E3. The principal controls are an echo-state representation, a matched-dimensional random nonlinear map, and a classical JL representation, all followed by the same finite Matérn readout grid. Raw-window ridge is retained as a low-cost reference; raw-window Matérn KRR is shown only when a predeclared resource guardrail is satisfied.

Q5: Which resource choices yield useful operating points? E5 performs one-factor sweeps over encoded width, reservoir count, observable locality, and contraction. Error is reported jointly with feature dimension, logical width, runtime, and memory so that non-monotonic behavior and Pareto frontiers remain visible.

Q6: How do finite shadows and stochastic channel trajectories affect prediction? E6 separates exact expectations, local-Pauli shadow sampling, reset-trajectory sampling, frozen-readout transfer, and noisy-feature training. Only the frozen-readout comparison is interpreted through corollary 13.

Q7: Does the architecture remain empirically useful outside the theorem's stochastic assumptions? E7 uses Lorenz–63 and Mackey–Glass as chronological forecasting stress tests. No stationarity or beta-mixing conclusion is inferred from them.

Q8: Do the controlled findings survive a limited external check? E8 evaluates three retained time-series regression datasets with different window lengths and channel counts. The study is a transfer sanity check, not a broad real-world benchmark.

Q9: Can a frozen exact-feature readout retain useful predictions when its inputs are replaced by IBM-QPU CSMoM features? E9 decomposes the hardware gap through an explicit comparison ladder: Aer exact expectations, Aer shadow sampling from the exact channel, local stochastic- trajectory CSMoM, and IBM Runtime CSMoM. It can establish compact operating- point feasibility and quantify perturbation, but not general hardware-noise robustness, scalability, or quantum advantage.

6.2 Common protocol

Outer and inner splits. Unless stated otherwise, theory-aligned synthetic experiments use five predeclared structural replicates, window length $w = 25$, stride $s = 100$, 5000 chronological outer-training windows, and 1000 chronological test windows. The final 1000 outer-training windows form the validation block; the first 4000 form the inner-training block. After selection, the chosen readout is refitted on all 5000 outer-training windows and the test set is evaluated once. Preprocessing is fitted on the relevant inner-training or outer-training features only. Forecasting targets may use future observations, but the model input ends at the declared prediction origin.

Reference QuaRK configuration. The reference simulation uses exact NVIDIA features with

$$n = 5, \qquad R = 3, \qquad k = 2, \qquad \lambda_r = 0.5 \text{ for all } r,$$

a ring topology, the tanh angle map, complex128 arithmetic, and all nonidentity Pauli strings of weight at most two. Its feature dimension is $p = 315$. The reference values are fixed protocol choices rather than claimed optima; E2 and E5 vary the relevant factors explicitly.

Readout selection. Features are standardized using the inner-training portion. For every representation and task, the finite Matérn KRR candidate set is

$$\nu \in \{1.5, 2.5, 5.0\}, \quad \xi \in \{0.25, 0.5, 1, 2, 4, 8\}, \quad \lambda_{\mathrm{K}} \in \{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^2\},$$

for $L = 90$ candidates. Validation MSE selects one candidate; ties within 10^{-12} are resolved lexicographically by $(\nu, \xi, \lambda_{\mathrm{K}})$. The corresponding objective parameter is $\eta = \lambda_{\mathrm{K}}/N_{\mathrm{tr}}$. Continuous optimization is not used in the main comparisons.

Randomness and pairing. Each structural replicate is generated from one root seed. Named child streams separately control data generation, split construction, JL projection, reservoir parameters, observable selection, reset trajectories, shadow bases, measurement outcomes, classical baselines, model selection, and repetitions. The same data, split, and task realizations are paired across methods. The complete seed bundle is stored with every artifact.

Primary metrics and uncertainty. For predictions $\widehat{y}_i$ and targets y_i, we report

$$\mathrm{MSE} = \frac{1}{m}\sum_{i=1}^{m}(\widehat{y}_i - y_i)^2, \qquad \mathrm{NRMSE}_{\mathrm{tr}} = \frac{\sqrt{\mathrm{MSE}}}{\widehat{\sigma}_{Y,\text{outer-train}}}.$$

Seed-level values are primary. Tables report mean and standard deviation; figures show paired structural replicates and a 95% percentile bootstrap interval over those replicates. With only five structural replicates, these intervals are descriptive and are not used as dichotomous significance tests. No test observation is used for selection.

Provenance and failures. Each run writes a versioned immutable artifact containing the resolved configuration, complete seed hierarchy, data and split checksums, program fingerprint, structured (N, R, K) features, selected candidate, predictions, metrics, timings, memory measurements, and execution status. Failed and incomplete runs are retained in the audit ledger. A run is excluded from an aggregate only for a predeclared integrity failure, unsupported capability, missing target, or unrecoverable backend failure; numerical underperformance is never an exclusion reason.

6.3 Experiment E1: Functional hierarchy and learning behavior

E1 uses the medium-persistence $d = 3$ VARMA family and the reference exact featurizer. A fixed $(\nu, \xi) = (2.5, 1)$ regularization path over $\lambda_{\mathrm{K}} \in \{10^{-8}, 10^{-7}, \ldots, 10^2\}$ records training, validation, and test MSE, RKHS norm, and the condition numbers of K and $K + \lambda_{\mathrm{K}} I$ for future projection, exponential memory, and Volterra memory. A nested-prefix study uses $N_{\mathrm{tr}} \in \{250, 500, 1000, 2000, 5000\}$, with the last 20% of each prefix used for selection and the same fixed test block. Delayed recall at $\tau \in \{1, 5, 10, 20\}$ and the cross-lag target at $(\tau_1, \tau_2) = (0, 15)$ complete the functional profile.

6.4 Experiment E2: Memory, persistence, and cross-family robustness

E2a uses low- and high-persistence VARMA realizations, delayed recall at $\tau \in \{1, 10, 20\}$, and $\lambda \in \{0.1, 0.3, 0.5, 0.8, 0.95\}$, while all other reference factors are fixed. Its primary outcome is the validation-selected contraction and the paired test NRMSE surface. E2b fixes the reference QuaRK configuration and uses low-, medium-, and high-persistence VARMA, HMM, and GARCH processes. The shared targets are

Table 2: Staged empirical design. Each stage isolates a distinct question; the full Cartesian product of generators, tasks, resources, and estimators is not run.

Stage	Retained setting	Primary evidence
E1	Reference VARMA; functional hierarchy	Capacity, conditioning, learning curves, memory diagnostics
E2	VARMA memory grid; VARMA/HMM/GARCH transfer	Persistence–contraction interaction and cross-family robustness
E3	Latent VARMA embedded to $d \leq 500$	Fixed encoded width, JL distortion, sparse/distributed failure modes
E4	Seven scenarios reused from E1–E3	Classical representation calibration under explicit matching rules
E5	Eleven one-factor QuaRK configurations	Error–resource Pareto structure
E6	Exact, shadow, and trajectory estimators	Feature and prediction degradation under finite sampling
E7	Lorenz–63 and Mackey–Glass	Out-of-scope chaotic forecasting stress tests
E8	Beijing PM2.5, live fuel moisture, hydraulic systems	Limited external transfer sanity check
E9	Two-qubit IBM CSMoM experiment	Compact hardware transfer and perturbation decomposition

exponential and Volterra memory; family-specific targets are true future projection, filtered-regime score, and next conditional volatility. QuaRK, ESN–Matérn, and matched-random-feature–Matérn runs from this stage are reused by E4 rather than recomputed.

6.5 Experiment E3: Ambient dimension and JL width control

E3 embeds one medium-persistence three-dimensional VARMA process into $d \in \{3, 30, 100, 300, 500\}$. In the sparse construction the first three coordinates contain the predictive process and the rest are bounded nuisance; in the distributed construction the same latent signal is spread through a fixed dense embedding before nuisance is added. The main curve fixes $n = 5$; additional width checks use $n \in \{3, 8\}$ at $d \in \{100, 500\}$. The exponential and Volterra teachers remain functions of the latent signal. Prediction is accompanied by projection time, feature time, peak memory, and

$$\widehat{\Delta}_{0.95} = \text{quantile}_{0.95} \left\{ \left| \frac{\|\Pi x - \Pi x'\|_2^2}{\|x - x'\|_2^2} - 1 \right| : (x, x') \in \mathcal{Q} \right\}, \tag{36}$$

where $\mathcal{Q}$ contains 10,000 predeclared nonidentical training-point pairs. The comparison includes QuaRK, classical JL–Matérn, matched random features, and raw ridge; raw Matérn is restricted to $d \leq 30$.

6.6 Experiment E4: Consolidated classical calibration

E4 introduces no new feature extraction. It aggregates seven predeclared scenarios: the three E1 reference tasks, medium-HMM regime filtering, medium-GARCH volatility prediction, and the $d = 500$ sparse and distributed Volterra settings. It reports QuaRK, ESN–Matérn, matched random features–Matérn, classical JL–Matérn where applicable, and raw ridge. Feature dimension, readout family, candidate count, runtime, and memory are reported separately; no single "matched budget" label is used.

6.7 Experiment E5: Resource and memory trade-offs

E5 uses medium-persistence VARMA with the Volterra and $\tau = 20$ recall targets. One-factor exact-feature sweeps around the reference configuration use

$$n \in \{3, 5, 8\}, \qquad R \in \{1, 3, 5\}, \qquad k \in \{1, 2, 3\}, \qquad \lambda \in \{0.1, 0.3, 0.5, 0.8, 0.95\}.$$

After deduplicating the shared reference point, this gives eleven configurations per task and replicate. The report includes NRMSE, feature dimension, encoded width, exact-feature runtime, peak GPU memory, and the analytical channel-update count NRw. Pareto membership is computed with respect to test NRMSE and measured feature-extraction runtime; it is reported descriptively, not used to select a test result.

6.8 Experiment E6: Finite-shadow and trajectory degradation

The principal E6 protocol trains one readout on exact reference features and evaluates it on CSMoM features for 256 predeclared test windows. It uses the exponential, Volterra, and $\tau = 10$ recall targets, $S \in \{100, 300, 1000, 3000, 10000\}$, ten MoM blocks, three structural replicates, and five independent shadow replicates. A 16-window Aer diagnostic at the same snapshot counts isolates shadow sampling from reset-trajectory sampling. A secondary noisy-feature-training study uses 512/128/256 train/validation/test windows, $S \in \{300, 1000, 3000\}$, three structural replicates, and three measurement replicates. Metrics include coordinate RMSE, per-example Euclidean feature error divided by $\sqrt{p}$, frozen-readout prediction perturbation, NRMSE, relative degradation from exact features, NRS state preparations, and $NRSw$ temporal channel applications.

6.9 Experiment E7: Out-of-scope chaotic stress tests

E7 follows the reduced matrix in Appendix F.3. Lorenz–63 uses forecast horizons $\{1, 5, 20\}$; Mackey–Glass uses delay parameters $\{17, 30\}$ and horizons $\{1, 10, 20\}$. Both use observation-noise levels $\{0, 0.02\}$, three trajectory replicates, chronological 3000/1000/1000 train/validation/test blocks, and a guard interval equal to the largest horizon. QuaRK is compared only with ESN–Matérn and matched random features–Matérn. We report NRMSE, persistence skill $1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{persistence}}$, and the first evaluated horizon at which skill is nonpositive.

6.10 Experiment E8: Limited real-world sanity checks

E8 uses the retained official splits for Beijing PM2.5, live fuel moisture, and hydraulic systems, specified in Appendix G.2. The final 20% of each official training split is used for inner validation. Five paired representation seeds are used for QuaRK, ESN, matched random features, and raw ridge. Raw Matérn is attempted only when its estimated Gram storage is below 2 GiB and a predeclared pilot completes within 30 minutes. Dataset-level MSE, training-normalized NRMSE, feature dimension, feature time, readout time, and peak memory are reported without average-rank or general-superiority claims.

6.11 Experiment E9: Compact IBM hardware validation

E9 uses a medium-persistence $d = 2$ VARMA process with $w = 4$, one two-qubit reservoir, $\lambda = 0.8$, and seven local observables. Exact NVIDIA features train two frozen readouts for $\tau = 1$ and $\tau = 3$ recall. The hardware subset contains eight predeclared test windows. CSMoM uses $S = 192$ snapshots and six MoM blocks. The same shadow bases and reset trajectories are reused in Aer and IBM-local comparisons and across two hardware executions separated by calibration time. The backend-selection, layout, job-chunking, failure, and provenance rules are fixed in Appendix H. The principal outcomes are feature RMSE relative to Aer exact, feature RMSE relative to the matched IBM-local realization, frozen-readout prediction perturbation and NRMSE, transpiled depth, two-qubit gate count, circuit and shot counts, and job completion. Measurement mitigation is not part of the primary protocol unless it is implemented and predeclared before the first hardware result is viewed.

Numerical-result status. This manuscript version fixes the protocol but does not state numerical conclusions for runs that have not yet produced checksum-validated artifacts. Every later result sentence, table entry, and figure will be generated from the versioned artifacts identified in Appendix C.

7 Discussion and Limitations

The analysis was designed to keep the dynamical, statistical, and measurement layers of QuaRK separate. This modularity clarifies what can be inferred from the current theory and where empirical evidence or additional analysis is still needed.

7.1 Statistical scope and memory trade-offs

The single-design risk bound is uniform over an RKHS norm ball after the projection, reservoirs, observable family, window scheme, kernel parameters, and regularization level have been fixed. Corollary 10 adds a controlled model-selection layer for a predeclared finite family, including sample-dependent validation selection, at a logarithmic cost in the number of candidates. Continuous optimization, adaptively enlarged search spaces, and sample-estimated preprocessing not included in that family still require separate analysis. The assumptions of stationary beta mixing and bounded labels are also substantive; nonstationarity, heavy-tailed targets, and online adaptation lie outside the present result.

The block length a and the window length w play different roles. Increasing a weakens the dependence penalty through $\beta_Z(as - w)$ but reduces the effective block count $\mu = N/(2a)$. Increasing w, or choosing larger contraction parameters, reduces the worst-case loss of remote history, but may increase circuit cost and retain information that is irrelevant to the task. The theory exposes these trade-offs; it does not select an optimal operating point.

7.2 Measurement, hardware, and end-to-end resources

The main learning theorem uses exact observable expectations. The classical-shadow result addresses only finite sampling under independent local Pauli measurements, and corollary 13 propagates this error through a fixed readout. Training with a noisy Gram matrix, coherent gate errors, reset imperfections, decoherence, calibration drift, and correlated measurements require separate treatment. Likewise, the circuit in Figure 2 is an ideal dilation of the replacement channel. The compact experiment in Section 6.11 instead uses pre-sampled reset trajectories and finite local-Pauli measurements. It can quantify transfer at one operating point, but it does not establish general robustness to hardware noise.

Resource accounting is therefore component-wise. The JL projection can reduce the logical encoded width on a finite point set, but applying a dense projection remains d-dependent. The ambient-dimension experiment in Section 6.5 is designed to test fixed-width behavior, empirical projection distortion, and predictive degradation as d grows; it cannot by itself establish end-to-end runtime independence or practical scalability. Sequential execution of the R reservoirs can avoid an R-fold increase in peak width, at the cost of more state preparations and runtime. With N windows and S snapshots per reservoir output, feature extraction requires NRS state preparations and $NRSw$ temporal channel applications. The shadow bound is logarithmic in the number of requested observables for fixed locality, but exponential in the locality k. Dense kernel ridge regression adds quadratic memory and cubic naive training time.

7.3 Comparative interpretation

The state update, reset mechanism, and observable measurement process are quantum, whereas the projection and Matérn readout are classical. None of the interpolation or generalization results identifies a predictive capability that is unavailable to a classical nonlinear reservoir or random feature map. Consequently, the practical value of the quantum featurizer must be assessed through matched comparisons that control feature dimension, readout, data splits, and model-selection effort. We do not claim quantum advantage, asymptotic speedup, state-of-the-art accuracy, or general superiority over classical temporal models.

The chaotic, real-world, and hardware experiments have evidential roles that differ from the theory-aligned synthetic study. Lorenz–63 and Mackey–Glass are explicit out-of-scope stress tests, while the three TSER datasets are limited external sanity checks for which stationarity and beta mixing are not asserted. The IBM experiment is a compact finite-shot feasibility check using a frozen readout. These studies may reveal transfer or failure, but they do not expand the formal scope of the risk theorem.

The method is general-purpose and is not tied to a sensitive deployment in its present form. Its foreseeable risks are the ordinary risks of temporal prediction—distribution shift, unreliable extrapolation, and overconfidence— together with overstated resource or advantage claims in an early-stage quantum setting. Making the analytical assumptions and resource ledger visible is intended to reduce these risks rather than to replace application-specific validation.

8 Conclusion

We introduced QuaRK, a temporal learning framework that combines a finite-set input projection, reset-contractive quantum reservoirs, concatenated local-observable features, and a regularized Matérn kernel readout. The architecture separates encoded width, reservoir memory, feature dimension, measurement cost, and classical kernel training, allowing each resource to be interpreted on its own terms.

The theoretical analysis mirrors this construction. Exact trace-norm contraction yields a unique causal state and a geometrically decaying finite-window feature error. Matérn RKHS stability transfers that error to predictions and squared risk, while strict positive definiteness characterizes the norm needed to interpolate distinct finite feature sets. For a fixed exact featurizer and stationary beta-mixing data, the independent-block analysis provides a one-sided risk bound governed by an effective block count and the finite-window approximation. A finite-family extension makes the guarantee simultaneous over a predeclared collection of reservoir and readout configurations, thereby covering validation-based selection with an explicit logarithmic penalty. A separate classical-shadow result connects observable locality and measurement accuracy to the evaluation of a fixed readout.

These results establish a coherent analytical basis for studying quantum reservoir kernels, while leaving comparative predictive value to empirical evaluation. The empirical protocol is organized around a predeclared hierarchy of forecasting, fading-memory, delayed-recall, cross-lag, regime-filtering, and volatility tasks; robustness across three beta-mixing data families; an ambient-dimension study of JL width control; matched classical and component controls; resource ablations; finite-shot behavior; out-of-scope chaotic stress tests; three limited real-world sanity checks; a compact IBM hardware transfer experiment; and uncertainty across repeated runs. Numerical conclusions will be added only after the corresponding artifacts have been audited.

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A Technical Background

A.1 Quantum states, replacement channels, and trace norm

An n-qubit density operator is a positive semidefinite matrix of trace one on $(\mathbb{C}^2)^{\otimes n}$. The trace norm is $\|A\|_1 = \mathrm{Tr}\sqrt{A^\dagger A}$, and the operator norm is $\|A\|_\infty$. We repeatedly use

$$|\mathrm{Tr}[AB]| \le \|A\|_\infty \|B\|_1 ,$$

unitary invariance of the trace norm, and the fact that two density operators have trace-norm distance at most two.

For completeness, the replacement map $\mathcal{R}_\sigma(A) = \mathrm{Tr}[A]\sigma$ is completely positive and trace preserving; this is a finite-dimensional instance of the operator-sum representation of quantum operations (Kraus, 1971). Indeed, if $\sigma = \sum_a r_a |a\rangle\langle a|$ and $\{|b\rangle\}_b$ is an orthonormal basis, the Kraus operators $K_{ab} = \sqrt{r_a}|a\rangle\langle b|$ satisfy

$$\sum_{a,b} K_{ab} A K_{ab}^\dagger = \mathrm{Tr}[A]\sigma, \qquad \sum_{a,b} K_{ab}^\dagger K_{ab} = I.$$

Thus the reset update in Equation (10) is a convex combination of two quantum channels.

A.2 Absolute regularity and the independent-block theorem

For sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, their absolute-regularity coefficient can be written as

$$\beta(\mathcal{A},\mathcal{B}) = \frac{1}{2} \sup_{\{A_i\},\{B_j\}} \sum_{i,j} |\mathbb{P}(A_i \cap B_j) - \mathbb{P}(A_i)\mathbb{P}(B_j)| ,$$

where the supremum ranges over finite measurable partitions from the two sigma-algebras. The coefficient is monotone under inclusion of either sigma-algebra, and measurable transformations cannot increase it (Bradley, 2005).

We use the following direct specialization of Mohri & Rostamizadeh (2008, Theorem 2). Let $(Q_j)_{j\in\mathbb{Z}}$ be stationary and beta mixing, let $\mathcal{F}$ be a fixed measurable class of functions into $[0,B]$, and let $m = 2\mu a$. Divide $Q_1,\dots,Q_m$ into 2μ consecutive blocks of length a, and form $\mathcal{S}_\mu$ by selecting the same fixed coordinate from every odd block. With the source paper's empirical Rademacher convention

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) := \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{f\in\mathcal{F}} \left| \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\widetilde{Q}_j) \right| \right], \tag{37}$$

if

$$\delta'_Q := \delta - 4(\mu-1)\beta_Q(a) > 0,$$

then, with probability at least $1-\delta$, simultaneously for every $f \in \mathcal{F}$,

$$\mathbb{E}[f(Q_0)] \le \frac{1}{m}\sum_{j=1}^{m} f(Q_j) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) + 3B\sqrt{\frac{\log(4/\delta'_Q)}{2\mu}}. \tag{38}$$

The theorem is one-sided. A two-sided statement would require an additional reverse-deviation argument and a corresponding confidence allocation. We use Equation (37) throughout so that the imported theorem and all constants in our specialization share exactly the same convention.

A.3 Spectral representation of the Matérn kernel

By Bochner's theorem (Bochner, 1933), the normalized stationary Matérn kernel has a probability spectral measure:

$$\kappa(z,z') = \mathbb{E}_\Omega\left[e^{\mathrm{i}\Omega^\top(z-z')}\right] = \mathbb{E}_\Omega\left[\cos(\Omega^\top(z-z'))\right]. \tag{39}$$

For the parameterization in Equation (11), its density on $\mathbb{R}^p$ is

$$p_{\nu,\xi}(\omega) = \frac{\Gamma(\nu + p/2)\,\xi^p}{\Gamma(\nu)(2\nu\pi)^{p/2}} \left(1 + \frac{\xi^2\,\|\omega\|_2^2}{2\nu}\right)^{-(\nu+p/2)}, \tag{40}$$

which is a centered multivariate Student density with 2ν degrees of freedom and scale matrix $\xi^{-2}I$; see Genton (2001); Porcu et al. (2024). It is strictly positive everywhere. For $\nu > 1$, its covariance exists and is

$$\mathbb{E}[\Omega\Omega^\top] = \frac{\nu}{(\nu-1)\xi^2} I. \tag{41}$$

These identities yield the section-stability constant used in lemma 4. Positivity of the density also yields strict positive definiteness on distinct Euclidean points, as made explicit in the proof of theorem 6; the general RKHS interpolation facts used there are standard (Aronszajn, 1950; Kimeldorf & Wahba, 1971).

A.4 Local Pauli classical shadows

In one snapshot of the local-Pauli shadow protocol, each qubit is measured in an independently and uniformly chosen X, Y, or Z basis. Inversion of the one-qubit measurement channel multiplies a nonidentity Pauli factor by three. Hence a Pauli string of weight q has squared shadow norm 3^q. The median-of-means guarantee of Huang et al. (2020) therefore gives simultaneous estimation of M nonidentity, weight-at-most-k Pauli expectations with $O(3^k\varepsilon^{-2}\log(M/\delta))$ independent snapshots. Identity observables, if included in a feature ledger, are known exactly and require no measurement.

A.5 Provenance of imported ingredients

The replacement-channel representation follows the standard operator-sum framework (Kraus, 1971), and the spectral representation of the stationary Matérn kernel invokes Bochner's theorem (Bochner, 1933). The finite-set projection result is the standard JL argument (Dasgupta & Gupta, 2003; Woodruff, 2014); the dependent-data inequality is imported from Mohri & Rostamizadeh (2008); the RKHS and representer facts are standard consequences of Aronszajn (1950); Kimeldorf & Wahba (1971); and the finite-union step uses the bounded-difference method of McDiarmid (1989) and standard Rademacher-complexity tools (Bartlett & Mendelson, 2002). The results specific to QuaRK are the way these ingredients are composed with its reset-contractive dynamics, finite-window approximation, observable map, and fixed-design temporal learning problem.

B Proofs

B.1 Finite-set JL projection

Proof. For a fixed nonzero vector v, concentration of $\|\Pi v\|_2^2 \,/\, \|v\|_2^2$ around one gives

$$\mathbb{P}\Big(\Big|\|\Pi v\|_2^2 - \|v\|_2^2\Big| > \varepsilon\,\|v\|_2^2\Big) \le 2e^{-c_0\varepsilon^2 n}$$

for a universal $c_0 > 0$. Apply this inequality to the fewer than $M^2/2$ nonzero differences $u - v$ and take a union bound. Choosing $n \ge C\varepsilon^{-2}\log(M/\delta)$ for a sufficiently large universal C makes the total failure probability at most δ. This is the standard finite-set JL argument (Dasgupta & Gupta, 2003; Woodruff, 2014). If $\mathcal{P}$ is random but independent of Π, the calculation holds conditionally on every realized $\mathcal{P}$, and hence also unconditionally. □

B.2 Trace-norm contraction

Proof. For density operators $\mathrm{Tr}[\rho] = \mathrm{Tr}[\rho'] = 1$, so the replacement components cancel when two outputs with the same input are subtracted:

$$T_{r,x}(\rho) - T_{r,x}(\rho') = \lambda_r V_r(x)(\rho - \rho')V_r(x)^\dagger.$$

Homogeneity and unitary invariance of the trace norm give Equation (15).

Iterating the equality for w common inputs yields

$$\left\|\rho_0^{(r)} - \rho_0^{\prime(r)}\right\|_1 \leq \lambda_r^w \left\|\rho_{-w}^{(r)} - \rho_{-w}^{\prime(r)}\right\|_1 \leq 2\lambda_r^w.$$

To construct the causal state, initialize the recursion at time $-m$ in an arbitrary density operator and call the resulting time-zero state $\rho_0^{[m]}$. If $m' > m$, regard the recursion started at $-m'$ as having some density operator at time $-m$. The last m updates are common, so

$$\left\|\rho_0^{[m']} - \rho_0^{[m]}\right\|_1 \leq 2\lambda_r^m.$$

Thus $(\rho_0^{[m]})_m$ is Cauchy in the finite-dimensional trace-norm space and converges. The bound also shows that the limit is independent of the chosen initial states. Shifting the argument in time produces a unique causal state sequence. For every finite start time, the time-zero state is a measurable function of a finite input block. Under Section 3.4, the causal state is their pointwise trace-norm limit and is therefore measurable. □

B.3 Finite-window feature error

Proof. For every r and $O \in \mathcal{O}$, lemma 2 and trace duality give

$$\left|\mathrm{Tr}\left[O\big(\rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)}\big)\right]\right| \leq \|O\|_\infty \left\|\rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)}\right\|_1 \leq 2\lambda_r^w.$$

Squaring and summing over r and O yields

$$\|\Phi_\infty - \Phi_w\|_2^2 \leq 4|\mathcal{O}| \sum_{r=1}^{R} \lambda_r^{2w}.$$

Taking square roots proves the first inequality. The second follows from $\lambda_r \leq \lambda_{\max}$. □

B.4 Matérn RKHS section stability

Proof. The reproducing property and normalization give

$$\begin{aligned}\|\kappa(z,\cdot) - \kappa(z',\cdot)\|_{\mathcal{H}_\kappa}^2 &= \kappa(z,z) + \kappa(z',z') - 2\kappa(z,z') \\ &= 2\left(1 - \varphi_{\nu,\xi}(\|z - z'\|_2)\right).\end{aligned}$$

Let $u = z - z'$. Using the spectral representation in Equation (39), the elementary inequality $1 - \cos t \leq t^2/2$, and Equation (41),

$$\begin{aligned}2(1 - \kappa(z,z')) &= 2\mathbb{E}[1 - \cos(\Omega^\top u)] \\ &\leq \mathbb{E}[(\Omega^\top u)^2] \\ &= u^\top \mathbb{E}[\Omega\Omega^\top] u \\ &= \frac{\nu}{(\nu-1)\xi^2} \|u\|_2^2.\end{aligned}$$

Taking square roots proves the claim. □

B.5 Prediction and loss truncation

Proof. For any $h \in \mathcal{H}_{\kappa,\Lambda}$, the reproducing property, lemma 4, and proposition 3 imply

$$\begin{aligned}|h(\Phi_\infty) - h(\Phi_w)| &= \left|\langle h, \kappa(\Phi_\infty,\cdot) - \kappa(\Phi_w,\cdot)\rangle_{\mathcal{H}_\kappa}\right| \\ &\leq \Lambda \|\kappa(\Phi_\infty,\cdot) - \kappa(\Phi_w,\cdot)\|_{\mathcal{H}_\kappa} \\ &\leq \frac{2\Lambda}{\xi}\sqrt{\frac{\nu|\mathcal{O}|}{\nu-1}\sum_{r=1}^{R}\lambda_r^{2w}}.\end{aligned}$$

This is Equation (20).

For $u, v \in [-\Lambda, \Lambda]$ and $|y| \le \Upsilon$,

$$\left|(u-y)^2-(v-y)^2\right| = |u-v|\,|u+v-2y| \le 2(\Lambda+\Upsilon)\,|u-v|\,.$$

Both predictions are in $[-\Lambda, \Lambda]$ by Equation (4). Apply the last display pointwise with $u = h(\Phi_\infty)$, $v = h(\Phi_w)$, use the prediction bound above, and take expectations. This gives $|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \le \tau_w$. □

B.6 Conditional interpolation

Proof. The Matérn spectral density in Equation (40) is strictly positive on $\mathbb{R}^p$. For any $\boldsymbol{c} \in \mathbb{R}^N$, Bochner's representation gives

$$\boldsymbol{c}^\top K \boldsymbol{c} = \int_{\mathbb{R}^p} \left| \sum_{i=1}^N c_i e^{\mathrm{i}\omega^\top z_i} \right|^2 p_{\nu,\xi}(\omega)\, d\omega.$$

Suppose that the z_i are distinct and $\boldsymbol{c} \ne 0$. Choose a direction $v \in \mathbb{R}^p$ outside the finite union of hyperplanes $\{v : v^\top(z_i - z_j) = 0\}$, so the scalars $v^\top z_i$ are pairwise distinct. The restriction $t \mapsto \sum_i c_i e^{\mathrm{i}tv^\top z_i}$ is a one-dimensional exponential polynomial with distinct frequencies and cannot vanish identically unless all $c_i = 0$. Thus the multivariate exponential polynomial is nonzero at some point and, by continuity, on an open set of positive Lebesgue measure. Since $p_{\nu,\xi} > 0$ everywhere, the integral is strictly positive. Hence K is positive definite and invertible.

Let $\boldsymbol{\alpha} = K^{-1}\boldsymbol{y}$ and

$$h_\star(\cdot) = \sum_{i=1}^N \alpha_i \kappa(z_i, \cdot).$$

Then $(h_\star(z_1), \ldots, h_\star(z_N))^\top = K\boldsymbol{\alpha} = \boldsymbol{y}$ and

$$\|h_\star\|^2_{\mathcal{H}_\kappa} = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha} = \boldsymbol{y}^\top K^{-1} \boldsymbol{y} = \Lambda_\star^2.$$

Moreover, if h is any other interpolant, then $g := h - h_\star$ satisfies $g(z_i) = 0$ for all i, so g is orthogonal to every kernel section $\kappa(z_i, \cdot)$ and hence to $h_\star$. Therefore $\|h\|^2_{\mathcal{H}_\kappa} = \|h_\star\|^2_{\mathcal{H}_\kappa} + \|g\|^2_{\mathcal{H}_\kappa}$, proving that $h_\star$ is the unique minimum-norm interpolant (Aronszajn, 1950; Kimeldorf & Wahba, 1971).

Thus $h_\star$ is feasible for every $\Lambda \ge \Lambda_\star$ and has zero empirical risk. For $0 \le \Lambda \le \Lambda_\star$, the function $h_\Lambda = (\Lambda/\Lambda_\star)h_\star$ is feasible and satisfies

$$h_\Lambda(z_i) - y_i = -\left(1 - \frac{\Lambda}{\Lambda_\star}\right) y_i.$$

Its empirical risk is the right-hand side of Equation (23); the minimum over the ball cannot be larger. The case $\boldsymbol{y} = 0$ is immediate. □

B.7 Mixing of the window process

Proof. By stationarity it is enough to compare the sigma-algebra generated by $(W_j)_{j \le 0}$ with that generated by $(W_j)_{j \ge a}$. Every raw variable occurring in a past window has time index at most zero, so

$$\sigma(W_j : j \le 0) \subseteq \sigma(Z_t : t \le 0).$$

The first raw input appearing in W_a has index $as - w + 1$, and all raw variables in later windows occur no earlier. Therefore

$$\sigma(W_j : j \ge a) \subseteq \sigma(Z_t : t \ge as - w + 1).$$

Monotonicity of $\beta(\cdot,\cdot)$ under sigma-algebra inclusion gives

$$\beta_W(a) \le \beta_Z(as - w + 1) \le \beta_Z(as - w),$$

where the last inequality uses that mixing coefficients are nonincreasing. For $a = 1$ and $s = w + g$, $as - w = g$. □

B.8 One-sided dependent-data risk bound

Proof. For $h \in \mathcal{H}_{\kappa,\Lambda}$, define

$$f_h(W_j) = \left(h(\Phi_w(X_{js-w+1:js})) - Y_{js}\right)^2.$$

By Equation (4) and $|Y_t| \le \Upsilon$,

$$0 \le f_h(W_j) \le B, \qquad B = (\Lambda + \Upsilon)^2.$$

The class $\mathcal{F}_\Lambda = \{f_h : h \in \mathcal{H}_{\kappa,\Lambda}\}$ is fixed independently of the sample under the analytical scope of Section 3.4. Apply Equation (38) to the stationary window process. By lemma 7,

$$\beta_W(a) \le \beta_Z(as - w),$$

Let $\delta'_W = \delta - 4(\mu - 1)\beta_W(a)$ denote the source-theorem remainder. The mixing bound gives $0 < \delta' \le \delta'_W$, so the source theorem applies; using the smaller δ' only enlarges the confidence term. Consequently, simultaneously for all $h \in \mathcal{H}_{\kappa,\Lambda}$,

$$\mathcal{R}_w(h) \le \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}}.$$

Finally, corollary 5 gives $\mathcal{R}_\infty(h) \le \mathcal{R}_w(h) + \tau_w$. Combining the two inequalities proves Equation (27). $\square$

B.9 Closed-form RKHS complexity bound

Proof. Condition on $\mathcal{S}_\mu = ((u_j, y_j))_{j=1}^{\mu}$, where $u_j = \Phi_w(x_j)$, and let $\mathcal{G}_\Lambda$ be the prediction class induced by $\mathcal{H}_{\kappa,\Lambda}$ on these features. Because the RKHS ball is symmetric,

$$\begin{aligned}
\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda) &= \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{\|h\|_{\mathcal{H}_\kappa} \le \Lambda} \left|\frac{2}{\mu}\left\langle h, \sum_{j=1}^{\mu} \sigma_j \kappa(u_j, \cdot)\right\rangle_{\mathcal{H}_\kappa}\right|\right] \\
&= \frac{2\Lambda}{\mu} \mathbb{E}_{\boldsymbol{\sigma}} \left\|\sum_{j=1}^{\mu} \sigma_j \kappa(u_j, \cdot)\right\|_{\mathcal{H}_\kappa} \\
&\le \frac{2\Lambda}{\mu}\sqrt{\sum_{j=1}^{\mu} \kappa(u_j, u_j)} = \frac{2\Lambda}{\sqrt{\mu}}.
\end{aligned}$$

For each j, define $\psi_j(t) = (t - y_j)^2 - y_j^2$. On $[-\Lambda, \Lambda]$, $\psi_j(0) = 0$ and ψ_j is $2(\Lambda + \Upsilon)$-Lipschitz. The absolute-value contraction principle (Bartlett & Mendelson, 2002) gives the safe bound

$$\mathbb{E}_{\boldsymbol{\sigma}} \sup_{h \in \mathcal{H}_{\kappa,\Lambda}} \left|\frac{2}{\mu}\sum_{j=1}^{\mu} \sigma_j \psi_j(h(u_j))\right| \le 4(\Lambda + \Upsilon)\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda) \le \frac{8\Lambda(\Lambda + \Upsilon)}{\sqrt{\mu}}.$$

The source convention contains an absolute value, so the label-only offset must also be retained. By Jensen's inequality and independence of the Rademacher signs,

$$\mathbb{E}_{\boldsymbol{\sigma}}\left|\frac{2}{\mu}\sum_{j=1}^{\mu} \sigma_j y_j^2\right| \le \frac{2}{\mu}\sqrt{\sum_{j=1}^{\mu} y_j^4} \le \frac{2\Upsilon^2}{\sqrt{\mu}}.$$

The triangle inequality therefore yields

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) \le \frac{8\Lambda(\Lambda + \Upsilon) + 2\Upsilon^2}{\sqrt{\mu}} = \frac{C_{\Lambda,\Upsilon}}{\sqrt{\mu}}.$$

Substitution into Equation (27), followed by the definition of τ_w, proves Equation (29). $\square$

B.10 Finite-family model selection

Proof. For candidate ℓ, let

$$\mathcal{F}_\ell := \left\{(x_{-w+1:0}, y) \mapsto (h(\Phi_w^{(\ell)}(x_{-w+1:0})) - y)^2 : h \in \mathcal{H}_\ell\right\},$$

and define the fixed union $\mathcal{F}_{\mathfrak{C}} := \bigcup_{\ell=1}^{L} \mathcal{F}_\ell$. Every function in this union is measurable and takes values in $[0, B_{\mathfrak{C}}]$. Apply Equation (38) once to $\mathcal{F}_{\mathfrak{C}}$, using lemma 7 exactly as in the proof of theorem 8. With probability at least $1-\delta$, simultaneously for every ℓ and $h \in \mathcal{H}_\ell$,

$$\mathcal{R}_{w,\ell}(h) \le \widehat{\mathcal{R}}_{N,w,\ell}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_{\mathfrak{C}}) + 3B_{\mathfrak{C}}\sqrt{\frac{\log(4/\delta')}{2\mu}}. \tag{42}$$

Condition on $\mathcal{S}_\mu$ and define

$$Z_\ell(\boldsymbol{\sigma}) := \sup_{f \in \mathcal{F}_\ell} \left| \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\widetilde{W}_j) \right|.$$

Changing one Rademacher sign changes Z_ℓ by at most $4B_{\mathfrak{C}}/\mu$. The bounded-difference exponential inequality (McDiarmid, 1989) therefore gives, for every $t > 0$,

$$\mathbb{E}_{\boldsymbol{\sigma}} \exp(t[Z_\ell - \mathbb{E}_{\boldsymbol{\sigma}} Z_\ell]) \le \exp\left(\frac{2t^2 B_{\mathfrak{C}}^2}{\mu}\right).$$

The log-sum-exp inequality then gives

$$\mathbb{E}_{\boldsymbol{\sigma}} \max_{\ell \in [L]} Z_\ell \le \max_{\ell \in [L]} \mathbb{E}_{\boldsymbol{\sigma}} Z_\ell + \frac{\log L}{t} + \frac{2tB_{\mathfrak{C}}^2}{\mu}.$$

Optimizing at $t = \sqrt{\mu \log L/(2B_{\mathfrak{C}}^2)}$ (with the $L = 1$ case immediate) yields the standard finite-union estimate (Bartlett & Mendelson, 2002)

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_{\mathfrak{C}}) \le \max_{\ell \in [L]} \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\ell) + 2B_{\mathfrak{C}}\sqrt{\frac{2\log L}{\mu}}.$$

The preceding closed-form calculation applies candidate by candidate, so

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\ell) \le \frac{8\Lambda_\ell(\Lambda_\ell + \Upsilon) + 2\Upsilon^2}{\sqrt{\mu}} \le \frac{A_{\mathfrak{C}}}{\sqrt{\mu}}.$$

Substitute these bounds into Equation (42) and apply the candidate-specific inequality $\mathcal{R}_{\infty,\ell}(h) \le \mathcal{R}_{w,\ell}(h) + \tau_{w,\ell}$. This proves Equation (30). Because the event is simultaneous over the complete fixed union, it remains valid after any measurable sample-dependent selector taking values in that union. $\square$

B.11 Geometrically mixing case

Proof. Since $\mu = N/(2a)$ and $a \ge 1$,

$$\begin{aligned} 4(\mu-1)\beta_Z(as-w) &\le \frac{2N}{a}\beta_0 e^{-c(as-w)} \\ &\le 2N\beta_0 e^{cw} e^{-cas}. \end{aligned}$$

The choice in Equation (31) makes the last expression at most $\delta/2$, so $\delta' \ge \delta/2$. It also gives $a = O(\log(N/\delta))$ with the remaining quantities fixed. Since $\mu = N/(2a)$, both stochastic terms in Equation (29) are $O(\sqrt{a/N}) = O(\sqrt{\log(N/\delta)/N})$.

For a nondivisible available sample size, retain $N_a = 2a\lfloor N/(2a)\rfloor$ observations. When $N_a \ge 2a$, stationarity permits applying the same bound to this retained consecutive segment, and the number discarded is at most $2a-1$. For fixed parameters, the chosen $a = O(\log N)$, so $a \le N/2$ and $N_a \ge 2a$ for all sufficiently large N. $\square$

B.12 Local-Pauli shadow requirement

Proof. For independently uniform local $X/Y/Z$ measurements, a nonidentity Pauli string of weight $q \le k$ has squared shadow norm $3^q \le 3^k$. The median-of-means classical-shadow guarantee applied to M observables therefore gives

$$S \ge C_{\mathrm{sh}} 3^k \varepsilon^{-2} \log(2M/\delta_{\mathrm{sh}})$$

for a universal constant C_{sh}; see Huang et al. (2020). For NR reservoir output states, assign failure probability $\delta_{\mathrm{sh}} = \delta_{\mathrm{tot}}/(NR)$ to each state and take a union bound. This gives Equation (33). Each state requires S independently randomized snapshots, for NRS state preparations. Since each snapshot replays the w-step recursion, the total is $NRSw$ channel applications. ☐

B.13 Finite-shot feature perturbations

Proof. The coordinate-wise event gives $\left\|\widehat{\Phi}_w - \Phi_w\right\|_2 \le \sqrt{R|\mathcal{O}|}\,\varepsilon$. The following argument is deterministic and uses only $\|h\|_{\mathcal{H}_\kappa} \le \Lambda$, so it holds simultaneously for every $h \in \mathcal{H}_{\kappa,\Lambda}$. The reproducing property and lemma 4 yield $\left|h(\widehat{\Phi}_w) - h(\Phi_w)\right| \le \Lambda \left\|\kappa(\widehat{\Phi}_w, \cdot) - \kappa(\Phi_w, \cdot)\right\|_{\mathcal{H}_\kappa} \le (\Lambda/\xi)\sqrt{\nu R|\mathcal{O}|/(\nu-1)}\,\varepsilon$, proving Equation (34). Both predictions lie in $[-\Lambda, \Lambda]$; the squared-loss Lipschitz inequality used in the proof of corollary 5 therefore gives Equation (35). ☐

C Implementation-Ready Experimental Protocol

This appendix is the execution contract for E1–E9. Values declared here are fixed before numerical results are inspected. A later implementation may change a value only through a versioned protocol revision that is recorded in the artifact manifest and applied to all affected methods.

C.1 Identifiers, roots, and artifact layout

The protocol version is `quark-empirical-v1`. Its structural replicate roots are

$$\mathcal{S}_{\mathrm{struct}} = \{1101, 1102, 1103, 1104, 1105\}.$$

Every root deterministically spawns the named streams `dataset_generation`, `dataset_split`, `jl_projection`, `reservoir_parameters`, `observable_selection`, `reset_trajectories`, `shadow_bases`, `measurement_outcomes`, `classical_baseline`, `model_selection`, and `replicate`. Measurement repetitions are child seed bundles of the structural root rather than integer offsets.

The canonical run identifier is

`<protocol>/<experiment>/<scenario>/<method>/<root>/<configuration-hash>`.

The corresponding directory is

```
storage/artifacts/experiments/quark-empirical-v1/
  <E1--E9>/<scenario>/<method>/root=<root>/<configuration-hash>/
```

A complete run contains the versioned feature artifact, data and split manifests, selected readout candidate, predictions, long-form metrics, timings, resource counters, environment, and status. Hardware runs additionally retain all Runtime job handles and raw shadow provenance. Aggregate files are written under `storage/artifacts/experiments/quark-empirical-v1/aggregate/`; they never replace the immutable run directories.

C.2 Common data, model, and readout objects

Synthetic outer split. For E1–E6, each process produces one ordered sequence of 6000 supervised windows after initialization or burn-in. Indices 0:3999, 4000:4999, and 5000:5999 are inner train, validation, and test, respectively. E1 learning curves use nested prefixes of the first 5000 windows and reserve the final 20% of each prefix for validation. No random shuffling is performed.

E1 Gaussian VARMA realization addendum. For the E1 reference process, the innovation covariance is I_3 and the bounded observation scale in Equation (46) is $c_x = 1$. Three Gaussian AR matrix directions are normalized to unit spectral norm, assigned positive normalized random weights, and multiplied by a common scalar found by a deterministic bracketed root solve so that the companion spectral radius is 0.7 to numerical tolerance 10^{-12}. The MA coefficients are $B_j = 0.5(0.5)^{j-1}V_j$, where the V_j are independent Gaussian directions normalized to unit spectral norm. The augmented ARMA transition covariance is obtained from the discrete Lyapunov equation and the initial augmented state is drawn from the resulting stationary Gaussian law. Thus E1 uses no burn-in approximation. Here and throughout the controlled protocol, $s = 100$ denotes the distance between consecutive window endpoints, not a gap added to w.

The named seed list is extended append-only with `task_functionals`; existing stream spawn keys are unchanged. E1's base directions u_3, v_3 and decay γ are fixed protocol constants, so this stream is recorded as unused for E1.

Reference program. The common reference program is

$$(n, R, k, \lambda, w, s) = (5, 3, 2, 0.5, 25, 100),$$

with all three reservoirs assigned the same reset rate, independent reservoir unitary draws, ring topology, Gaussian JL projection, tanh angle map, and every nonidentity Pauli word of weight at most two. The observable order is lexicographic within weight and qubit support and is stored explicitly. The resulting (N, R, K) tensor has $K = 105$ and flattens to $p = 315$ in reservoir-major, observable-minor order.

Finite readout grid. For standardized features, the common grid is

$$\mathcal{G}_{\mathrm{KRR}} = \{1.5, 2.5, 5.0\} \times \{0.25, 0.5, 1, 2, 4, 8\} \times \{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}\}, \tag{43}$$

where the entries are $(\nu, \xi, \lambda_{\mathrm{K}})$. The feature scaler and validation score are fitted on inner train and validation only. The lexicographically first minimizer is selected and refitted on outer train. A method whose representation is stochastic regenerates only the representation stream declared by the protocol; it does not receive a larger readout search.

Principal controls. The controlled comparison uses:

1. `quark-exact`: the exact NVIDIA QuaRK representation;
2. `esn-matern`: a tanh ESN whose state dimension equals the reported QuaRK feature dimension, with spectral radius $\{0.5, 0.9\}$, input scale $\{0.2, 1.0\}$, and leak rate $\{0.5, 1.0\}$, followed by Equation (43);
3. `random-map-matern`: a fixed Gaussian random affine map of the flattened standardized window to p coordinates followed by coordinatewise tanh and the same readout grid;
4. `jl-matern`: the same $d \to n$ projection applied at every time point, flattened over the window, followed by the same readout;
5. `raw-ridge`: standardized flattened windows with ridge $\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}, 10^{4}\}$; and
6. `raw-matern`: standardized flattened windows and the common Matérn grid, only under the stated resource guardrail.

The ESN candidate count is reported separately from the readout count. No method is described as compute matched unless time, memory, and candidate count are all shown.

C.3 Metric definitions

Let m be the number of examples in the evaluated split and let $p = RK$. In addition to MSE and training-normalized NRMSE from Section 6.2, the following quantities are retained.

Paired error difference. For method A, control B, and structural root s,

$$\Delta^{(s)}_{A-B} = \mathrm{NRMSE}^{(s)}_{A} - \mathrm{NRMSE}^{(s)}_{B}.$$

Negative values favor A. Pairing is valid only when data, split, and task streams match.

Feature errors. For estimated and exact flattened features $\widehat{\phi}_i, \phi_i \in \mathbb{R}^p$,

$$\mathrm{FRMSE}_{\mathrm{coord}} = \sqrt{\frac{1}{mp}\sum_{i=1}^{m}\left\|\widehat{\phi}_i - \phi_i\right\|_2^2}, \qquad e_{2,i} = \frac{\left\|\widehat{\phi}_i - \phi_i\right\|_2}{\sqrt{p}}.$$

The report gives the mean, median, and 95th percentile of $e_{2,i}$. For a frozen readout h, prediction perturbation is

$$\Delta_h = \frac{1}{m}\sum_{i=1}^{m}\left|h(\widehat{\phi}_i) - h(\phi_i)\right|.$$

Relative degradation is

$$D_{\mathrm{rel}} = \frac{\mathrm{NRMSE}_{\mathrm{estimated}} - \mathrm{NRMSE}_{\mathrm{exact}}}{\max(\mathrm{NRMSE}_{\mathrm{exact}}, 10^{-12})}.$$

Readout diagnostics. For dual coefficients $\boldsymbol{\alpha}$, the fitted RKHS norm is $\sqrt{\boldsymbol{\alpha}^\top K \boldsymbol{\alpha}}$. Condition numbers are computed in float64 as $\kappa_2(K)$ and $\kappa_2(K + \lambda_{\mathrm{K}} I)$; a singular value below $10^{-14}\sigma_{\max}$ is reported as numerical singularity rather than silently clipped.

JL diagnostic. The pair set $\mathcal{Q}$ contains 10,000 unordered nonidentical training point pairs generated once from the `model_selection` stream. In addition to $\widehat{\Delta}_{0.95}$, E3 reports the fraction whose relative squared-distance distortion exceeds 0.5. Pairs with raw squared distance below 10^{-12} are excluded and replaced deterministically.

Compute and resource metrics. Wall-clock stages are projection, feature compilation, feature execution, feature transfer, readout selection, and final fit/prediction. GPU timings are synchronized. Peak device memory is obtained from the GPU memory pool before cleanup; host peak memory uses process resident-set high-water mark. We also store n, physical qubits when applicable, p, NR reservoir outputs, NRw exact temporal channel updates, NRS shadow snapshots/state preparations, and $NRSw$ sampled temporal updates. Hardware runs additionally store circuit groups, submitted PUBs, total shots, completed and missing results, transpiled depths, two-qubit gate counts, layouts, queue time, and execution timestamps.

C.4 Experimental master matrix

Table 3: Implementation master matrix. Feature runs may serve several tasks because the same structured feature tensor is reused by multiple readouts.

ID	Data and task combinations	Varied factors	Replication and estimator	Principal outputs
E1	Medium VARMA; future, exponential, Volterra, delays $1, 5, 10, 20$, cross-lag $(0, 15)$	KRR path; train prefixes $250, 500, 1000, 2000, 5000$	5 roots; exact NVIDIA; one cached feature tensor/root	loss path, RKHS norm, conditioning, learning and delay curves
E2a	Low/high VARMA; delays $1, 10, 20$	$\lambda = 0.1, 0.3, 0.5, 0.8, 0.95$	5 roots; 50 exact feature runs	contraction–persistence NRMSE surface
E2b	VARMA/HMM/GARCH at low/medium/high persistence; shared and family-specific targets	family and persistence only	5 roots; 45 QuaRK feature runs plus ESN/random controls	cross-family absolute and paired errors
E3	Latent VARMA in sparse/distributed embeddings; exponential and Volterra	$d = 3, 30, 100, 300, 500$; $n = 3, 5, 8$ only at stated points	5 roots; 90 QuaRK feature runs plus controls	error, JL distortion, time, memory
E4	Seven selected E1–E3 scenarios	method only	artifact aggregation; no new features	calibrated comparison table
E5	Medium VARMA; Volterra and delay 20	one-factor n, R, k, λ sweeps	5 roots; 55 exact feature runs	Pareto error, time, memory, width, dimension
E6	Medium VARMA; exponential, Volterra, delay 10	snapshots and estimator protocol	3 structural roots; independent measurement replicates	feature and prediction degradation
E7	Lorenz–63 and Mackey–Glass forecasting	horizon, observation noise, MG delay	3 trajectory roots; exact NVIDIA and two controls	NRMSE and persistence skill
E8	Beijing PM2.5, live fuel moisture, hydraulic systems	method only	5 representation roots; exact NVIDIA and controls	dataset-level transfer table
E9	$d = n = 2, w = 4, R = 1$ VARMA; delays 1 and 3	execution layer only	8 windows; $S = 192, G = 6$; 2 hardware periods	feature/prediction ladder and hardware provenance

C.5 Experiment-specific execution contracts

C.5.1 E1: `quark-e1-functional-v1`

The feature artifact is generated once per structural root with `ExactFeatureEstimator(complex128)` and the reference NVIDIA program. Targets are stored independently from the feature artifact. The regularization path uses fixed $(\nu, \xi) = (2.5, 1)$ and the eleven powers $10^{-8}, \ldots, 10^{2}$. Prefix results are valid only if nested-index checks pass. A task with zero outer-training standard deviation is marked invalid before fitting. The main display is a three-panel figure: regularization path, learning curve, and delayed-recall profile. Full path tables and conditioning values remain in the appendix.

C.5.2 E2: `quark-e2-memory-v1` and `quark-e2-family-v1`

E2a generates separate feature tensors for every $(\text{persistence}, \lambda, \text{root})$ triple and reuses each tensor for three delays. E2b creates one tensor for every $(\text{family}, \text{persistence}, \text{root})$ combination. HMM states and GARCH conditional variances are written only to teacher-target artifacts; they never enter model inputs. Generator stationarity checks and persistence parameters are mandatory manifest fields. The main display combines a contraction–delay heat map with a cross-family paired-error panel.

C.5.3 E3: `quark-e3-dimension-v1`

A three-dimensional latent VARMA path is generated first. Sparse and distributed embeddings are then deterministic functions of the latent path and the named nuisance/embedding streams. Thus all dimensions for one root share the same latent realization. Main fixed-width runs use $n = 5$ for all five dimensions. Width sensitivity adds $n = 3, 8$ only for $d = 100, 500$; the $n = 5$ values are reused. Raw Matérn is skipped, with status `guardrail-not-run`, for $d > 30$. The main display has prediction and JL-distortion panels; detailed timing and memory tables are supplementary.

C.5.4 E4: `quark-e4-calibration-v1`

The aggregator fails if any expected scenario lacks at least four paired structural roots. Its seven rows are: E1 future, E1 exponential, E1 Volterra, medium-HMM regime, medium-GARCH volatility, $d = 500$ sparse Volterra, and $d = 500$ distributed Volterra. Columns are method NRMSE mean/std, paired difference to QuaRK, feature dimension, candidate count, feature time, and peak memory. Missing raw-Matérn guardrail runs appear as em dashes with a status footnote, not as failures or wins.

C.5.5 E5: `quark-e5-resources-v1`

The unique program list is generated by changing one field of the reference program and deduplicating by program fingerprint. All programs use the same data and readout grid. Runtime is measured after one unreported warm-up and three timed executions of the same compiled feature plan; the median is stored. A run that exhausts device memory is retained with its requested dimensions, last measured allocation, and `resource-exhausted` status. Pareto membership uses median feature time and mean test NRMSE and is recomputed from stored rows.

C.5.6 E6: `quark-e6-shadows-v1`

For the frozen-readout study, 256 test windows are selected before measurement by equally spaced indices from the full test block. Snapshot counts are $100, 300, 1000, 3000, 10000$, with ten MoM blocks, three structural roots, and five measurement replicates. The exact scaler and readout remain frozen. A 16-window Aer subset, using the same bases and outcome seeds wherever semantically possible, estimates shadow-only error from an exact channel. The secondary noisy-training study uses 512/128/256 train/validation/test windows, $S = 300, 1000, 3000$, three structural roots, and three measurement replicates; training and evaluation use independent measurement-outcome child streams. Raw snapshots are retained for the Aer subset and one audit replicate per snapshot count, not for every large GPU run.

C.5.7 E7: `quark-e7-chaos-v1`

After burn-in, each chaotic trajectory is standardized using the inner-training block and bounded by a fixed tanh map before projection. Input windows use stride one. A guard interval of 20 observations separates validation from test so that no future target crosses a split boundary. Horizon-specific readouts reuse the same features. Numerical integration failure, nonfinite state, or invariant-range escape is an integrity failure and is retained in the ledger. The complete matrix is in Appendix F.3.

C.5.8 E8: `quark-e8-real-v1`

Official train/test indices are immutable. The final 20% of official train is validation. Dataset inputs use the archive's retained window representation; training-only standardization is applied channelwise before JL projection. Raw Matérn is attempted only if estimated float64 Gram storage is below 2 GiB and a single readout-grid pilot completes within 30 minutes. The aggregate requires all five paired representation roots for each stochastic method, but no dataset is dropped because one method underperforms. Dataset specifications are in Table 7.

C.5.9 E9: `quark-e9-ibm-v1`

The exact execution ladder, backend-selection score, fixed-layout rule, Runtime-job chunking, and missing-result policy are given in Appendix H. The protocol submits no more than 360 unique circuit groups and 1536 total shadow shots per hardware period. Any provider limit lower than the full group count triggers deterministic chunking by the ordered group key; it does not change the structural plan. Hardware results are reported for every completed period, including an unsuccessful or partial period, and are never pooled without the calibration timestamps.

C.6 Main-paper and appendix display contract

The main paper is limited to six empirical displays after results are available:

1. **Figure 3: Functional and sample-complexity profile** (E1): regularization, nested-sample, and delay panels;
2. **Figure 4: Memory and dependence map** (E2): contraction–delay heat map and cross-family paired errors;
3. **Figure 5: Fixed-width ambient-dimension study** (E3): NRMSE and JL distortion against d;
4. **Table 2: Classical calibration** (E4): the seven retained scenarios and transparent matching fields;
5. **Figure 6: Resource and measurement operating points** (E5–E6): error–runtime Pareto panel and frozen-readout shot curve; and
6. **Figure 7: Compact IBM transfer ladder** (E9): feature error and frozen-readout degradation, accompanied by one small hardware-resource table.

E7 and E8 are summarized in text and shown in the appendix unless their results change a central conclusion.

The supplementary displays include all seed-level metric tables, full KRR paths, all E2 persistence levels, all E3 width and timing rows, complete E5 sweeps including failures, E6 shadow-only and noisy-training diagnostics, chaos horizon/noise panels, dataset-level E8 predictions, and the complete E9 job, calibration, layout, depth, count, and missing-result ledger. Negative and failed runs are retained rather than hidden.

C.7 Run completion and audit rules

A scenario is complete only when all expected manifests pass checksum and ordering validation, all selected readouts can be reproduced from stored features, and every displayed aggregate maps to immutable run identifiers. Resumable hardware jobs are incomplete until decoded results or a terminal provider status are stored. When fewer than the required replicates finish, the result is shown as incomplete and no mean-only replacement is reported. All plotting and table scripts consume long-form aggregate files and are forbidden from reading ad hoc notebook variables.

D Theory-Aligned Dependent Synthetic Families

The controlled experiments use synthetic processes for which stationarity, beta mixing, and bounded labels can be justified from the data-generating mechanism. The three families below deliberately represent different sources of temporal dependence: linear vector dynamics, latent regime persistence, and conditional heteroskedasticity. They are not intended to exhaust the processes covered by theorem 8; rather, they provide a compact set of qualitatively distinct and analytically traceable test beds.

D.1 Finite-block supervised transformations

We first record a closure property used for all three constructions. It separates the mixing argument for the input process from the choice of bounded supervised target.

Lemma 14 (Finite-block transformations preserve beta mixing; cf. Bradley, 2005)**.** *Let $(V_t)_{t\in\mathbb{Z}}$ be a stationary beta-mixing process, let $(E_t)_{t\in\mathbb{Z}}$ be an i.i.d. process independent of V, and let ψ and f be measurable maps. For integers $\ell_-, \ell_+ \geq 0$, define*

$$X_t = \psi(V_t), \qquad Y_t = f(V_{t-\ell_-:t+\ell_+}, E_t), \qquad Z_t = (X_t, Y_t).$$

Then Z is stationary and, for every $q > \ell_- + \ell_+$,

$$\beta_Z(q) \leq \beta_V(q - \ell_- - \ell_+). \tag{44}$$

If $|f| \leq \Upsilon$, then the bounded-label condition in Section 3.4 also holds.

Proof. Let $U_t = (V_t, E_t)$. Because E is i.i.d. and independent of the complete process V, adjoining the two disjoint innovation sigma-algebras does not increase the absolute-regularity coefficient; this is the standard independent augmentation property of beta mixing (Bradley, 2005). The past sigma-algebra of Z is contained in $\sigma(U_t : t \le \ell_+)$, while its future sigma-algebra at lag q is contained in $\sigma(U_t : t \ge q - \ell_-)$. Their raw-time separation is $q - \ell_- - \ell_+$. Monotonicity under sigma-algebra inclusion and measurable maps therefore yields

$$\beta_Z(q) \le \beta_U(q - \ell_- - \ell_+) = \beta_V(q - \ell_- - \ell_+).$$

Stationarity follows because the joint process (V, E) is stationary and the construction is shift equivariant. $\square$

The complete controlled task suite is defined separately in Appendix E. All shared tasks are bounded finite-block maps of the underlying process, with one future step used only for the genuine forecasting target. The HMM filtering score and GARCH conditional-volatility target are also bounded measurable functions of the corresponding finite-dimensional state constructions. Their compatibility with the assumptions of theorem 8 is summarized in lemma 18.

D.2 Stable vector VARMA

Let (U_t) satisfy the vector VARMA(p, q) recursion

$$U_t = \sum_{i=1}^{p} A_i U_{t-i} + \varepsilon_t + \sum_{j=1}^{q} B_j \varepsilon_{t-j}, \tag{45}$$

where the innovations are i.i.d. in $\mathbb{R}^d$. The input exposed to the learner is the bounded coordinatewise transformation

$$X_t = \tanh(U_t / c_x), \tag{46}$$

with a fixed scale $c_x > 0$.

Lemma 15 (Stable vector ARMA processes are geometrically beta mixing; Mokkadem, 1988)**.** *Assume that the autoregressive polynomial $A(z) = I - \sum_{i=1}^{p} A_i z^i$ is nonsingular for every $|z| \le 1$, that the innovations have an absolutely continuous distribution, and that $\mathbb{E} \log^+(1 + \|\varepsilon_0\|_2) < \infty$. Then the standard causal VARMA series is well defined and stationary, and the geometric complete-regularity result of Mokkadem (1988) gives*

$$\beta_U(q) \le C_{\mathrm{V}} \rho_{\mathrm{V}}^q$$

for constants $C_{\mathrm{V}} < \infty$ and $\rho_{\mathrm{V}} \in (0, 1)$. Consequently, the bounded process in Equation (46), together with the bounded task maps in Appendix E, satisfies the stochastic assumptions of theorem 8, up to the finite lag shift in Equation (44).

The moment condition is mild and is satisfied by the Gaussian and Student innovations considered in the implementation. The mixing conclusion is the geometric complete-regularity result of Mokkadem (1988); the final statement follows from the data-processing property of beta mixing (Bradley, 2005) and lemma 14.

Realization used in the controlled suite. We retain the existing $d = 3$, $p = q = 3$ generator with centered Gaussian innovations. For every saved configuration, the spectral radius of the companion matrix is checked to be below one and recorded with the generated artifact. The medium-dependence operating point is rescaled to companion spectral radius 0.7; additional sensitivity points use 0.4 and 0.9. The same fixed matrices, innovation covariance, scaling c_x, and target vectors are reused across all compared methods for a given seed.

High-dimensional extension. The ambient-dimension experiment does not simulate a dense d-dimensional VARMA recursion. Instead, one medium-persistence latent process $L_t \in \mathbb{R}^3$ is generated and reused across $d \in \{3, 30, 100, 300, 500\}$. In the sparse construction,

$$X_t^{(d)} = \tanh\left((L_t^\top, \zeta_t^\top)^\top\right),$$

where $\zeta_t \in \mathbb{R}^{d-3}$ is independent bounded nuisance. In the distributed construction,

$$X_t^{(d)} = \tanh(A_d L_t + \sigma_\zeta \zeta_t),$$

where $A_d \in \mathbb{R}^{d\times 3}$ has orthonormal columns, is sampled once per structural root, and $\sigma_\zeta = 0.1$. The target remains a function of L_t, so increasing d changes the observation representation rather than the teacher. All dimensions for one root share the same latent path and nested nuisance streams. This construction isolates fixed-width compression and avoids conflating ambient dimension with an $O(d^2)$ dense VARMA simulation.

D.3 Finite-state hidden Markov observations

Let (S_t) be a Markov chain on $[K] = \{1,\dots,K\}$ with transition matrix P. Conditional on $S_t = j$, draw an emission

$$U_t = \mu_j + L_j \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d), \qquad X_t = \tanh(U_t), \tag{47}$$

where the emission noises are independent of the latent chain. This is the standard finite-state hidden Markov construction of Baum & Petrie (1966).

Lemma 16 (Positive finite-state hidden Markov models are geometrically beta mixing; Dobrushin, 1956; Bradley, 2005)**.** *Suppose that S_0 has the stationary distribution of P and*

$$\epsilon_P := \min_{i,j\in[K]} P_{ij} > 0.$$

Then the latent chain and the observed process in Equation (47) are stationary and geometrically beta mixing. In particular,

$$\beta_X(q) \le \beta_S(q) \le (1 - K\epsilon_P)^q. \tag{48}$$

After adjoining the bounded shared or filtered-regime targets in Appendix E, the resulting supervised process satisfies the assumptions of theorem 8 with the lag adjustment from lemma 14.

Proof. The Dobrushin coefficient satisfies

$$\delta(P) := \sup_{i,i'} \|P(i,\cdot) - P(i',\cdot)\|_{\mathrm{TV}} \le 1 - K\epsilon_P.$$

For a stationary finite-state Markov chain, $\beta_S(q) = \sum_i \pi_i \|P^q(i,\cdot) - \pi\|_{\mathrm{TV}}$. Dobrushin contraction therefore gives $\beta_S(q) \le \delta(P)^q \le (1 - K\epsilon_P)^q$ (Dobrushin, 1956; Bradley, 2005). Write the emission as $X_t = g(S_t, E_t)$ with i.i.d. E_t independent of S. Independent augmentation and measurable data processing give $\beta_X(q) \le \beta_{(S,E)}(q) = \beta_S(q)$. The supervised conclusion then follows from lemma 14. $\square$

Realization used in the controlled suite. We use $K = d = 3$, initialize the latent chain from its stationary (discrete-uniform) distribution, and set

$$P_\varrho = \varrho I_3 + \frac{1-\varrho}{2}(\mathbf{1}\mathbf{1}^\top - I_3), \qquad \varrho \in \{0.60, 0.80, 0.95\}. \tag{49}$$

The three emission means and covariances are

$$\mu_1 = (-1.5, 0, 1.0)^\top, \quad \mu_2 = (0.5, 1.5, -1.0)^\top, \quad \mu_3 = (1.5, -1.0, 0.5)^\top, \qquad L_j = 0.35 I_3.$$

The middle persistence level $\varrho = 0.80$ is the main operating point; 0.60 and 0.95 test faster and slower regime switching. Because all entries of P_ϱ are positive, every realization satisfies lemma 16.

D.4 Gaussian GARCH(1,1)

To study volatility clustering, let

$$R_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \omega + \alpha_{\mathrm{G}} R_{t-1}^2 + \beta_{\mathrm{G}} \sigma_{t-1}^2, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{50}$$

which is the GARCH(1,1) model introduced by Bollerslev (1986). We expose three bounded return-derived coordinates to the learner,

$$X_t = \left(\tanh(R_t/2), \tanh(|R_t|/2), \tanh(R_t^2/4)\right)^\top. \tag{51}$$

The observation map exposes signed return, magnitude, and squared-return information while keeping the learner input bounded. The family-specific prediction target is the conditional-volatility state in Equation (63).

Lemma 17 (Gaussian GARCH(1,1) is geometrically beta mixing under a moment contraction; Bougerol & Picard, 1992; Carrasco & Chen, 2002)**.** *Assume $\omega > 0$, $\alpha_{\mathrm{G}} > 0$, $\beta_{\mathrm{G}} \geq 0$, Gaussian innovations, and that for some $s > 0$,*

$$\mathbb{E}\left[(\alpha_{\mathrm{G}} \varepsilon_0^2 + \beta_{\mathrm{G}})^s\right] < 1. \tag{52}$$

Then the stationarity conditions of Bougerol & Picard (1992) and the Gaussian-density regularity conditions used by Carrasco & Chen (2002) are satisfied: the GARCH state has a unique strictly stationary solution and is geometrically beta mixing. Hence the bounded observation process in Equation (51), together with the shared tasks and the conditional-volatility target in Appendix E, satisfies the stochastic assumptions of theorem 8 after the lag adjustment in lemma 14.

Condition Equation (52) is a convenient sufficient specialization of the cited stationarity and mixing results. It implies $\mathbb{E} \log(\alpha_{\mathrm{G}} \varepsilon_0^2 + \beta_{\mathrm{G}}) < 0$ by Jensen's inequality. In the Gaussian realization, the transparent condition $\alpha_{\mathrm{G}} + \beta_{\mathrm{G}} < 1$ implies Equation (52) with $s = 1$.

Realization used in the controlled suite. We set $\omega = 1 - \alpha_{\mathrm{G}} - \beta_{\mathrm{G}}$, so the stationary second moment is one, and use

$$(\alpha_{\mathrm{G}}, \beta_{\mathrm{G}}) \in \{(0.10, 0.60), (0.10, 0.80), (0.07, 0.90)\}. \tag{53}$$

These settings have persistence sums 0.70, 0.90, and 0.97, respectively; the middle setting is used in the main cross-family comparison. The other two provide low- and high-persistence sensitivity checks. All three satisfy lemma 17 with $s = 1$.

D.5 Common dataset realization and empirical role

The cross-family robustness suite uses the same windowing and split protocol for all three generators. For each predeclared data-generating seed, we form $N_{\mathrm{tr}} = 5000$ training windows and $N_{\mathrm{te}} = 1000$ test windows with $w = 25$ and stride $s = 100$. The first 5000 windows form the training portion and the final 1000 form a chronological test portion; the final 20% of the training portion is reserved for inner validation. The five structural roots are $1101, 1102, 1103, 1104, 1105$, and all methods share the corresponding data streams. HMM trajectories start exactly in stationarity. For the Gaussian VARMA realization, we use its linear state-space representation, solve the associated discrete Lyapunov equation, and draw the augmented initial state from the resulting stationary Gaussian law. For GARCH, the theorem concerns the ideal stationary solution; the numerical generator approximates that law using an independent burn-in of 10^5 raw time steps. This approximation is kept identical across methods and is reported as a simulation approximation, not as proof of exact stationarity.

Table 4: Controlled dependent-family suite. The middle persistence setting is used in the main comparison; the remaining settings probe sensitivity to the strength of temporal dependence.

Family	Dependence mechanism	Persistence settings	Main targets
VARMA	Linear vector recurrence	companion radius in $\{0.4, 0.7, 0.9\}$	shared suite; latent high-dimensional embeddings
HMM	Latent regime switching	$\varrho \in \{0.60, 0.80, 0.95\}$	exponential, Volterra, filtered regime
GARCH	Conditional variance recursion	$\alpha_G + \beta_G \in \{0.70, 0.90, 0.97\}$	exponential, Volterra, conditional volatility

The detailed functional, architecture, high-dimensional, and shot studies remain concentrated on VARMA to avoid multiplying every experiment by three. HMM and GARCH are used for the cross-family experiment with the same selected QuaRK configuration, the shared exponential and Volterra tasks, one family-specific task, and the principal matched classical controls. The exact allocation is given in Table 5.

E Empirical Temporal Functionals

The controlled experiments separate the dynamics that generate the inputs from the temporal map that generates the response. This appendix defines the task family used throughout the synthetic study and fixes its normalization, delays, and allocation across experiments. The hierarchy follows the usual reservoir-computing distinction between linear memory and nonlinear information processing (Jaeger, 2001; Dambre et al., 2012), while the exponentially weighted and Volterra-type tasks are finite-window versions of classical fading-memory operators (Boyd & Chua, 1985).

E.1 Notation and common normalization

Let

$$\boldsymbol{X}_{t,w} := (X_{t-w+1}, \ldots, X_t) \in ([-1,1]^d)^w$$

be the input window presented to the learner. Choose directions $u_d, v_d \in \mathbb{R}^d$ satisfying $\|u_d\|_1, \|v_d\|_1 \leq 1$, and fix the memory-decay parameter $\gamma = 0.8$. For the base dimension $d = 3$, the common directions are

$$u_3 = \frac{1}{3}(1, -1, 1)^\top, \qquad v_3 = \frac{1}{3}(1, 1, -1)^\top.$$

All task directions, delays, decay parameters, and any task-generation seeds are fixed before the data are simulated; all methods evaluated on a given dataset receive the same realized task. Define

$$A_w := \sum_{j=0}^{w-1} \gamma^j = \frac{1-\gamma^w}{1-\gamma}. \tag{54}$$

The normalizations below place every controlled target in $[-1, 1]$, so the bounded-label constant can be taken as $\Upsilon = 1$ for this suite.

E.2 Shared temporal tasks

One-step future projection. The genuine forecasting task uses one observation beyond the input window,

$$F_{\text{fore}}(\boldsymbol{X}_{t,w}, X_{t+1}; u_d) := u_d^\top X_{t+1}. \tag{55}$$

The learner receives observations only through time t; X_{t+1} is used solely to construct the label. This distinguishes the task from an instantaneous decoding check based on X_t.

Exponentially weighted linear memory. Let

$$L_u(t) := \sum_{j=0}^{w-1} \gamma^j u_d^\top X_{t-j}. \tag{56}$$

The normalized fading-memory target is

$$F_{\mathrm{exp}}(\boldsymbol{X}_{t,w}; u_d) := \frac{L_u(t)}{A_w}. \tag{57}$$

It requires information from the full window while assigning greater weight to recent observations.

Second-order Volterra-type memory. With $L_v(t)$ defined as in Equation (56) using v_d, set

$$F_{\mathrm{vol}}(\boldsymbol{X}_{t,w}; u_d, v_d) := \frac{L_u(t) + \frac{1}{2} L_v(t)^2}{A_w + \frac{1}{2} A_w^2}. \tag{58}$$

This task combines distributed linear memory with a smooth nonlinear interaction across lags.

Delayed recall. For a delay $\tau \in \{1, 5, 10, 20\} \subset \{0, \ldots, w-1\}$, define

$$F_{\mathrm{delay},\tau}(\boldsymbol{X}_{t,w}; u_d) := u_d^\top X_{t-\tau}. \tag{59}$$

The resulting delay curve is a direct diagnostic of how much information about a particular past observation remains in the final reservoir feature vector. It is interpreted as an empirical memory profile, not as an estimator of the worst-case contraction bound.

Sparse cross-lag interaction. For the predeclared delays $(\tau_1, \tau_2) = (0, 15)$, define

$$F_{\mathrm{cross}}(\boldsymbol{X}_{t,w}; u_d, v_d) := \left(u_d^\top X_{t-\tau_1}\right)\left(v_d^\top X_{t-\tau_2}\right). \tag{60}$$

Unlike the dense Volterra target, this functional isolates one nonlinear interaction between a recent and a distant observation.

E.3 Family-specific tasks

The common tasks make performance comparable across generators. Two additional targets exploit the defining latent structure of the HMM and GARCH families.

HMM filtered regime score. For the three-state HMM in Appendix D.3, assign state scores $a = (-1, 0, 1)^\top$. Using the known generator parameters, compute the finite-window filtering probabilities

$$\pi_{t,w}(j) := \mathbb{P}(S_t = j \mid X_{t-w+1:t}), \qquad j \in \{1, 2, 3\}, \tag{61}$$

by the standard forward recursion, initialized from the stationary state probabilities at the left edge of the window (Baum & Petrie, 1966). The target is the posterior regime score

$$F_{\mathrm{HMM}}(\boldsymbol{X}_{t,w}) := \sum_{j=1}^{3} a_j \pi_{t,w}(j). \tag{62}$$

The learner receives only the emission window; neither the latent state nor the filtering probabilities are included in its input.

GARCH conditional-volatility state. For the GARCH process in Appendix D.4, let $\bar{\sigma}^2 := \omega/(1 - \alpha_{\mathrm{G}} - \beta_{\mathrm{G}})$ be the stationary second moment under the finite-variance parameterization (Bollerslev, 1986). The bounded target

$$F_{\mathrm{GARCH},t} := \tanh\left[c_\sigma \log\left(\frac{\sigma_{t+1}^2}{\bar{\sigma}^2}\right)\right], \qquad c_\sigma = \frac{1}{2}, \tag{63}$$

asks the learner to recover the next conditional-volatility state from the bounded return-derived input window. The latent variance is used only to construct the synthetic label.

E.4 High-dimensional observation constructions

The high-dimensional experiment changes the observation map while keeping the three-dimensional teacher fixed. Let $L_t \in [-1, 1]^3$ be the bounded latent VARMA process. For every ambient dimension, the exponential and Volterra targets are computed from $L_{t-w+1:t}$ using u_3 and v_3. The learner instead receives either the sparse or distributed observation $X_t^{(d)}$ defined in Appendix D.2. In the sparse construction, the informative coordinates are the first three and all remaining coordinates are nuisance. In the distributed construction, the informative subspace is spread through A_d, which is generated from the named JL-independent embedding stream and stored with the dataset artifact. Thus label range and intrinsic task difficulty do not increase merely because d increases.

This design distinguishes two possible compression failures. Sparse signal may be diluted by indiscriminate projection when only a few coordinates matter; distributed signal asks whether a global low-dimensional geometry survives the same fixed encoded width. No target direction is selected after inspecting prediction results.

E.5 Implementation contract for target generation

A target generator receives a window ending at time t and an immutable context object. Window-only functionals use only the window. The genuine forecast functional reads X_{t+1} from the context, the HMM score reads the precomputed finite-window filtering probabilities, and the GARCH target reads σ^2_{t+1}. These fields are teacher-only and are excluded from the model input and preprocessing. The context, target parameters, and teacher arrays are checksum-validated and stored separately from feature artifacts.

The implementation must not label $u^\top X_t$ as one-step forecasting. Current-point projection may be retained as an optional smoke test under the identifier `current-point-decoding`, but it is not an E1 forecasting result. Delayed-recall and cross-lag targets are deterministic window-only objects, while all direction vectors are created once per dataset artifact and reused for every example and method.

E.6 Assumption compatibility

Lemma 18 (The controlled functional suite satisfies the bounded-label and mixing requirements)**.** *For each process realization in Appendix D, all targets in Equations (55), (57) to (60), (62) and (63) are measurable and bounded by one in absolute value. When paired with the corresponding input process, the resulting supervised process is stationary and beta mixing under the conditions of lemmas 15 to 17.*

Proof. The ℓ_1-normalization of the directions and $X_t \in [-1, 1]^d$ give the bounds for forecasting, delayed recall, and the cross-lag product. Equation Equation (54) and the denominators in Equations (57) and (58) give the same bound for the two fading-memory targets. The HMM score is a convex combination of values in $[-1, 1]$, and the GARCH target is bounded by the outer hyperbolic tangent. Each common task is a finite-block function of the underlying process. The forecast target uses one future observation and therefore invokes lemma 14 with $\ell_+ = 1$; the remaining shared tasks have $\ell_+ = 0$. The HMM score is a measurable function of a finite emission block. Although it is indexed by $t+1$, $\sigma^2_{t+1} = \omega + \alpha_{\mathrm{G}} R_t^2 + \beta_{\mathrm{G}} \sigma_t^2$ is a measurable function of the current stationary GARCH state. The conclusion follows from lemma 14 and the family-specific mixing lemmas. No unbounded label noise is added in the theory-aligned controlled suite. □

E.7 Task allocation across the empirical study

The full Cartesian product of generators and targets is intentionally avoided. Table 5 assigns each functional to a specific scientific question.

Table 5: Compact allocation of controlled functionals. "Shared" means that the same mathematical target is applied to each listed generator; the family-specific targets use latent variables only to construct labels.

Experiment	Generators	Functionals
Functional hierarchy and learning curves	VARMA, medium persistence	future projection, exponential memory, Volterra, delay curve, cross-lag
Memory and persistence study	VARMA at low/high persistence	delayed recall at $\tau \in \{1, 10, 20\}$ under a contraction sweep
Cross-family robustness	VARMA, HMM, GARCH at low/medium/high persistence	shared exponential and Volterra tasks; VARMA future projection; HMM filtered score; GARCH volatility state
Ambient-dimension study	latent VARMA observed at $d \in \{3, 30, 100, 300, 500\}$	latent exponential and Volterra teachers under sparse/distributed observations
Resource and memory ablations	VARMA, medium persistence	Volterra and delayed recall at $\tau = 20$
Finite-shot study	VARMA, medium persistence	exponential memory, Volterra, and delayed recall at $\tau = 10$
Chaotic stress tests	Lorenz–63 and Mackey–Glass	multi-horizon forecasting and partial-observation prediction; see Appendix F
Real-world sanity checks	Beijing PM2.5, live fuel moisture, hydraulic systems	archive-provided scalar regression targets; see Appendix G
IBM hardware feasibility	compact $d = 2$ VARMA subset	delayed recall at $\tau \in \{1, 3\}$; see Appendix H

F Out-of-Scope Chaotic Stress Tests

The chaotic experiments are deliberately outside the stationary beta-mixing analysis. Their purpose is to test forecasting behavior when nearby states can separate rapidly, not to extend theorem 8. All numerical integration choices, splits, horizons, and failure rules are fixed before model results are inspected.

F.1 Lorenz–63

We integrate the classical Lorenz system (Lorenz, 1963)

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z, \tag{64}$$

with $(\sigma, \rho, \beta) = (10, 28, 8/3)$, fourth-order Runge–Kutta step 10^{-3}, and stored sampling interval $\Delta t = 0.05$. A burn-in of 1000 stored samples is discarded. Each of three initial-condition roots is a small predeclared perturbation of $(1, 1, 1)$; the perturbation vector and the complete integrated checksum are stored.

The learner receives all three standardized coordinates, mapped through a fixed coordinatewise tanh. We use window length $w = 25$, stride one, observation-noise standard deviation $\sigma_{\text{obs}} \in \{0, 0.02\}$, and horizons $h \in \{1, 5, 20\}$. The primary target is the normalized future x-coordinate. A second partial-observation task gives the learner only (x, y) and predicts future z. Noise is added before training-only standardization and uses a named stream shared across methods.

F.2 Mackey–Glass

We integrate the delay differential equation (Mackey & Glass, 1977)

$$\dot{x}(t) = \frac{\beta x(t - \tau)}{1 + x(t - \tau)^n} - \gamma x(t), \qquad (\beta, \gamma, n) = (0.2, 0.1, 10), \tag{65}$$

with delay $\tau \in \{17, 30\}$, integration step 0.1, unit stored sampling interval, and a constant history perturbed by the trajectory seed. We discard 5000 stored samples as burn-in. Inputs are training-standardized and bounded through the same tanh transformation as Lorenz. Window length is 25, observation noise is $\{0, 0.02\}$, and forecast horizons are $h \in \{1, 10, 20\}$.

F.3 Chronological protocol and metrics

Each trajectory provides 3000 inner-training, 1000 validation, and 1000 test windows. A guard interval of 20 raw observations separates validation and test origins so no future target crosses the boundary. Three trajectory roots are used. The reference exact QuaRK program, ESN–Matérn, and matched random features–Matérn share the readout grid in Equation (43). Horizon-specific labels reuse the same feature artifact.

The report gives training-normalized NRMSE and persistence skill

$$\mathrm{Skill}_{\mathrm{pers}} = 1 - \frac{\mathrm{MSE}_{\mathrm{model}}}{\mathrm{MSE}_{\mathrm{persistence}}},$$

where the persistence prediction repeats the latest available target coordinate. The failure horizon is the first evaluated horizon with nonpositive mean skill; if none occurs it is reported as "beyond evaluated range." Integration failure, nonfinite values, or failure to reproduce the stored trajectory checksum is an integrity failure rather than a model result.

Table 6: Reduced chaotic stress-test matrix.

System	Structural settings	Targets	Compared representations
Lorenz–63	$\Delta t = 0.05$, noise $\{0, 0.02\}$, horizons $\{1, 5, 20\}$	future x; future z from (x, y)	QuaRK, ESN, matched random map
Mackey–Glass	$\tau \in \{17, 30\}$, noise $\{0, 0.02\}$, horizons $\{1, 10, 20\}$	future scalar state	QuaRK, ESN, matched random map

G Limited Real-World Sanity Checks

The controlled synthetic suite is the primary empirical contribution. E8 adds three retained time-series extrinsic-regression datasets only to test whether qualitative conclusions transfer beyond designed generators. Stationarity, beta mixing, and bounded-label assumptions are not asserted for these data.

G.1 Retained datasets

Table 7: Real-world sanity-check datasets and retained official splits.

Dataset	Train	Test	w	d	Diagnostic role
Beijing PM2.5	5000	2000	24	9	Moderate multivariate environmental series and short context.
Live Fuel Moisture Content	3000	1000	365	7	Long seasonal context with a modest channel count.
Hydraulic Systems	1159	290	60	728	Very high-dimensional multivariate record testing the fixed-width projection path.

The archive-provided scalar regression targets and original train/test splits are retained (Tan et al., 2021). No synthetic target or latent state is introduced. The final 20% of each official training split is the validation block. Channelwise centering and scaling are fitted on inner train and recomputed on full official train after selection. Missing-value handling, window order, and dataset checksum are stored in the dataset manifest.

G.2 Comparison and guardrails

The comparison uses the exact NVIDIA QuaRK representation, ESN–Matérn, matched random features–Matérn, and raw ridge. Five paired representation roots are used. The common finite Matérn grid and validation metric are shared. Raw-window Matérn is attempted only when the estimated float64 Gram storage is below 2 GiB and one predeclared pilot candidate completes within 30 minutes; otherwise it is recorded as `guardrail-not-run`. The test set is evaluated once after representation and readout selection.

Primary metrics are MSE and training-standard-deviation-normalized RMSE. We also report feature dimension, projection time, feature-extraction time, readout time, and peak host/device memory. Dataset-level seed values and paired differences are retained. No average rank or win count is used to turn three datasets into a general superiority statement.

G.3 Interpretation boundary

A favorable E8 result is evidence that a controlled finding can survive on the corresponding dataset; it is not evidence of broad real-world robustness or practical deployment. An unfavorable result is retained because it identifies a transfer failure. The complete predictions and all method failures are provided in the appendix artifacts.

H Compact IBM Hardware Validation

E9 asks a narrow question: can a readout fitted from exact features retain useful predictions when the same eight inputs are represented by finite-shot, stochastic-trajectory features from an IBM QPU? The study is not designed to establish hardware-noise robustness across devices, quantum advantage, or scalable deployment.

H.1 Program, data, and frozen readout

The hardware-compatible program uses a medium-persistence $d = 2$ VARMA process, window length $w = 4$, stride $s = 100$, one two-qubit reservoir, ring/chain edge $(0, 1)$, $\lambda = 0.8$, the tanh angle map, and

$$\mathcal{O}_{\mathrm{HW}} = \{X_0, Y_0, Z_0, X_1, Y_1, Z_1, Z_0Z_1\}.$$

The stored observable labels are `IX,IY,IZ,XI,YI,ZI,ZZ` in the package's qubit-order convention. The two tasks are delayed recall at $\tau = 1$ and $\tau = 3$. One exact NVIDIA feature artifact is generated for 2000 inner-training, 500 validation, and 500 test windows. Two Matérn readouts are selected with Equation (43), refitted on the 2500 non-test windows, and then frozen. The hardware subset consists of eight equally spaced indices from the 500-window test block; those indices are stored before any hardware job is submitted.

The CSMoM estimator uses $S = 192$ local-Pauli snapshots and six deterministic MoM blocks. The shadow-basis and reset-trajectory arrays are created from independent named streams and reused by the local circuit and hardware layers. Measurement outcomes are independent across executions. The feature scaler and readout are always those fitted from exact features.

H.2 Comparison ladder

For the same eight windows and program, E9 computes:

1. **Aer exact**: analytic expectations from the independent $2n + 1$-qubit density-matrix dilation; this is the mathematical feature reference for the subset;
2. **Aer shadow-only**: CSMoM samples drawn from the exact Aer channel, isolating local-Pauli finite-shadow error;
3. **IBM-local trajectories**: the IBM circuit compiler and pre-sampled last-reset trajectories executed on an ideal local Aer sampler, combining trajectory and shadow sampling without device noise; and

4. **IBM hardware**: the identical structural plan transpiled and executed through IBM Runtime.

The distinction is semantic: Aer exact is exact mixed-channel evolution; Aer shadow-only estimates that exact state; IBM-local and hardware use one pre-sampled reset trajectory per shadow snapshot. No simulator-only noise instruction is described as a hardware run.

H.3 Backend and physical-layout policy

At the first hardware period, the driver lists operational non-simulator IBM backends with at least two qubits. For every connected physical pair (q_0, q_1), it computes

$$s(q_0, q_1) = e_{2\mathrm{q}}(q_0, q_1) + \frac{0.25}{2}\big(e_{\mathrm{ro}}(q_0) + e_{\mathrm{ro}}(q_1)\big),$$

where unavailable calibration values make the pair ineligible. The pair with smallest score is selected; scores within 5% are broken by backend pending-job count and then lexicographically by backend and qubit IDs. Backend-properties payload, selection candidates, score, and selection time are persisted.

Transpilation uses optimization level one, the named transpiler seed, and the selected pair as a fixed initial layout. No post-result remapping is allowed. The same backend and physical pair are reused in a second period at least 24 hours later when operational. If they are unavailable, the replacement is selected by the same rule and is reported as a separate operating point rather than pooled with the first.

H.4 Circuit grouping, Runtime jobs, and cost bound

Each snapshot is represented by its window, reservoir, sampled last-reset suffix, and local measurement basis. Snapshots with identical tuples share one circuit and are executed as repeated shots. With $N_{\mathrm{HW}} = 8$, $w = 4$, and $n = 2$, at most

$$N_{\mathrm{HW}}(w + 1)3^n = 360$$

unique groups are possible, while the total primary shots remain $N_{\mathrm{HW}}S = 1536$ per period. Ordered groups are split into chunks of at most 100 Runtime PUBs, giving at most four jobs per period. Each job handle, ordered group range, and expected snapshot count is written before waiting for results, so decoding is resumable.

A failed chunk is retried once with the same circuits and structural plan. There is no imputation. A period contributes feature and prediction aggregates only if all 1536 expected snapshots decode successfully. Partial periods remain in the job-success table with their completed counts. The primary design uses at most 3072 QPU shots over two periods; the single-retry contingency raises the hard bound to 4608 shots. This bound excludes local simulator and transpilation work.

H.5 Mitigation and calibration provenance

The primary E9 protocol uses no measurement mitigation because the current hardware adapter does not expose a predeclared mitigation implementation. Mitigation may be added only as a versioned secondary experiment implemented and tested before any mitigated hardware result is inspected. It cannot replace the raw-hardware result.

For every transpiled circuit and job, the artifact records backend name, backend-properties update time, submission and completion times, physical layout, depth, two-qubit gate count, total operation counts, Runtime job ID, shots, expected and decoded snapshots, and terminal status. Calibration payloads are stored as raw JSON-compatible snapshots where provider policy permits, with a checksum otherwise.

H.6 Metrics, uncertainty, and displays

For every non-exact ladder layer, we report coordinate feature RMSE and the per-window $e_{2,i}$ relative to Aer exact. Hardware is additionally compared with the matched IBM-local CSMoM realization, which shares bases and trajectories but not outcomes. The frozen-readout report includes mean absolute prediction perturbation, MSE, NRMSE, and relative degradation for both delays. Uncertainty is summarized by a snapshot bootstrap that resamples complete snapshot indices within each window and reconstructs CSMoM with six blocks; the two calibration periods are displayed separately rather than treated as i.i.d. replicates. Per-window paired differences are shown alongside the bootstrap interval.

The main paper receives one compact two-panel figure: feature error across the four ladder layers and exact-versus-hardware frozen-readout predictions for the eight windows. A small adjacent table reports backend, period, layout, median and maximum depth, total two-qubit gates, circuit groups, shots, and completion status. The appendix retains the full per-observable bias, per-window errors, transpilation rows, calibration metadata, and job ledger. A null or degraded result is scientifically valid: it identifies the hardware operating point at which exact-feature transfer fails.